

lumina-par — Design Document

ai-ee v1 pipeline

generated 2026-08-07 13:20:56

Contents

1	Board and Run Metadata	1
2	Overview	1
3	Requirements	1
4	Architecture	12
5	Schematic	24
6	Layout	31
7	Verification	31
8	DFM and Fabrication	31
9	Run Record (Appendix)	31
10	Artifact Index	35

1 Board and Run Metadata

Field	Value
board	lumina-par
workspace	boards/lumina-par
phase	P4
gate status	all 1 recorded gates pass
state created	2026-07-28T15:00:15
state updated	2026-08-07T13:20:24

2 Overview

3 Requirements

Requirements: lumina-par (LUM-DTR-PAR-A / LUM-PAR-A)

Sources, in precedence order (later overrides earlier): 1. `brief/00-lumina-system-context.md` - shared system context. 2. `brief/03-rgbw-par-daughter-brief.md` - this board's brief. 3. `brief/01-carrier-board-brief.md` - the mating carrier, CONTEXT ONLY. 4. `brief/05-lumina-closed-decisions.md` - BINDING; supersedes the open-decision table in 00 s6 and corrects figures in 00 and 03. 5. `brief/06-connector-icd.md` - FROZEN interface control document (ICD-01), owned by LUM-CAR-A. Hard input. Never redefined here.

Unstated low-risk items are marked **ASSUMED**: inline. Section 9 holds every unknown that changes the design or touches safety. Section 8 flags one >30 V condition that is unconditional (48 V is on the connector whether or not this board taps it) and one conditional >3 A condition.

Two figures in the briefs are dead and are NOT used anywhere below: the “~8.5 W sustained” daughter budget in 00 s5.1 and 03 PAR-REQ-11, and the Si3402-B / AN956 “~10 W regulated” it derives from. The carrier uses a TPS2378-class PD controller plus a 100 V buck; the re-derived budget is **8.6-9.3 W (af) / 18.7-20.0 W (at)**. PAR-REQ-11 itself still holds hard.

0. Formal amendments (owner decision at H1, 2026-07-28)

These requirements were **changed by decision at the H1 checkpoint**, not quietly failed. The original text is quoted in full so the record shows exactly what was altered and why. Both amendments follow from the H1-Q4 decision: ****same-reel sourcing, with the EEPROM fitted but shipped empty****, because no colour-measuring instrument exists for this build today.

AMD-01 - PAR-REQ-16 changes from a binning specification to a same-reel sourcing requirement > **Original (brief ‘03’ s4, PAR-REQ-16)**: “LED binning specified in the BOM for > all four channels. Unbinned parts across 8 fixtures will not satisfy > PAR-REQ-06.”

Amended to: *All emitters for the whole build shall be procured in a single transaction from a single reel per part number, and the reel/date code shall be recorded in the build record. A chromaticity or flux bin shall be specified in the BOM wherever the vendor publishes one; where the vendor publishes none, the same-reel requirement stands in its place.*

Why it could not be met literally: LCSC/JLC does not generally expose a chromaticity or flux bin as an orderable attribute, and the selected emitter vendor publishes no bin data at all (`research/led-emitter.md` s9). A BOM line demanding a bin that cannot be ordered is unbuildable. Parts from one reel are same-batch and closely matched in practice, which is the outcome PAR-REQ-16 was written to obtain. ****This is a genuine reduction in guarantee, not an equivalent substitution:**** same-reel is an empirical correlation, whereas a published bin is a contractual limit. PAR-REQ-06 (consistent colour across fixtures) is therefore carried at **reduced confidence** and its verification moves to a bench comparison of the assembled fixtures.

AMD-02 - PAR-REQ-17’s calibration data is deferred, not the hardware > **Original (brief ‘03’ s4, PAR-REQ-17)**: “Provide for per-fixture calibration: > measured per-channel scaling stored in the daughter board’s ID EEPROM > (CAR-REQ-07, EEPROM option) so firmware can normalise fixtures to each other. > If the resistor-divider ID option is chosen instead, calibration has to live on > the host, and that trade must be made deliberately.”

Amended to: *The board shall fit the ID divider bottom leg AND a 24C32 I2C EEPROM. The EEPROM ships **empty**: per-channel scaling coefficients are not measured or populated for the first build. The provision is retained so that calibration can be added later without a respin.*

Why: the premise that EEPROM and divider are alternatives is not what ICD-01 says - s2/s3.3 has them coexisting, the divider carrying board type and the EEPROM per-unit calibration, which is why the second dedicated ID pin was deleted. So the “deliberate trade” the original text asks for does not arise; both are fitted. What is genuinely deferred is the **data**, because there is no colorimeter or spectrometer in this project today (H1-P10). ****Consequence to carry:**** until the EEPROM is populated, fixture-to-fixture matching rests entirely on AMD-01’s same-reel sourcing.

1. Function

The colour and mood workhorse of the LUMINA fixture set: an RGBW par daughter board that stacks above the universal LUM-CAR-A carrier on the frozen ICD-01 expansion connector and turns four (possibly five) carrier PWM channels into four independent constant-current LED drives. It runs continuously rather than in bursts, so its problem is not peak power but quality of dimming and consistency between the 6-8 fixtures in the room: visually stepless dimming at low output, slow hue

drift over tens of seconds with no banding, and no per-fixture tint variation in a synchronised full-room wash. It carries LED drivers, thermal sensing, an independent over-temperature shutdown, the ENABLE gate, the board ID leg and the calibration store - and nothing else. It has no MCU, no network interface, no external connector of any kind, and no energy store for burst output (that is the strobe board's problem, not this one's).

2. Interfaces

Every interface on this board is through ICD-01. There are no others. ICD-01 s9 is explicit: a daughter may not add an external connector of any kind - no barrel jack, no DMX, no second Ethernet. Ethernet on the carrier is the fixture's only external connection, and that is what makes the non-isolated PoE topology compliant.

2.1 J3 - POWER block, 2x7, daughter side

- Part: **CONNFLY DS1023-27SF11*** socket (14 pos, 2.54 mm, 600 V, 3 A/contact, 8.5 mm body). Second sources on the same land pattern: HanElectricity 2541FV, Boomele 2.54-2*7P, HCTL HC-PZ254. JLC Extended - there is no JLC Basic part in any board-to-board family, so "prefer Basic" is unachievable here and is not a selection error.
- **Reverse-mounted on the daughter's BOTTOM side, facing down** (ICD s7.3).
- Pin map (ICD s3.1), authoritative, do not re-derive: 1/3/5 +48V_SW; 2/4/6/7/8/10/13 GND; 9/11 +12V; 12/14 +3V3. Position 1 silkscreen-marked with a triangle.
- Rail order along the connector is 48 -> GND -> 12 -> 3.3 so that no single-position mis-seat can put a higher rail on a lower rail's pin.

2.2 J4 - SIGNAL block, 2x12, daughter side

- Part: **CONNFLY DS1023-212SF11*** socket (24 pos, 600 V, 3 A/contact, 8.5 mm body), same mounting rules and same second-source situation.
- Pin map (ICD s3.2), authoritative: PWM0..7 at positions 1,2,5,6,7,8,11,12; GND at 3,4,9,10,13; DSPI_SCK 14, MOSI 15, MISO 16, CSn 17; I2C_SCL 18, SDA 19; ADC0 20, ADC1 21; ID_ADC 22; ENABLE 23; FAULT 24. **No 48 V exists anywhere on this connector.**

2.3 Signal contract (ICD s3.3) - what this board must honour

Signal	Requirement on this board
'PWM0..7'	3.3 V push-pull, **13-bit at 9.766 kHz default** (CAR-REQ-12). PWM0-3 = LEDC timer 0, PWM4-7 = timer 1; channels on a timer share frequency AND resolution. 'ASSUMED:' R,G,B,W map to PWM0-3 (timer 0); a 5th channel (question 4) takes PWM4 on timer 1, programmed identically. LEDC timers 2 and 3 are unallocated and available on request - that is the only sanctioned route to a different PWM frequency for this daughter, and it needs the carrier owner's agreement, not a unilateral change.
'I2C_SCL/SDA'	400 kHz open drain. **Pull-ups are on the carrier (4.7 k). This board must NOT fit its own.** The daughter owns the whole address space; the carrier reserves nothing - so EEPROM and any digital temp sensor must be allocated non-colliding addresses here.
'ADC0', 'ADC1'	daughter -> carrier, 0-3.3 V, **source impedance <= 10 kohm**. This is the PAR-REQ-12 thermistor path; an NTC divider must be sized to meet the 10 kohm ceiling at every temperature of interest.

```

| 'ID_ADC' | The carrier fits the TOP leg (10 k to +3V3); **this board fits the
BOTTOM leg to GND**. The code value is allocated by the carrier owner, not
chosen here - see question 15. |
| 'ENABLE' | carrier -> daughter, push-pull, **active HIGH**. This board must
fit its own **100 kohm pull-down**, gate **every** output stage with it (driver
EN pins, any gate driver, any charge path), and **never latch it locally**. A
carrier PWM pin can glitch ~60 us at power-up; ENABLE is what makes that a
no-op. |
| 'FAULT' | daughter -> carrier, **open drain, active low**, 10 k pull-up on
the carrier, wire-OR'd with the carrier's eFuse fault. **Never drive it
high.** |
| 'DSPI_*' | Available, <= 26 MHz, mode 0, shared bus with one CS. Not expected
to be needed by a 4-channel PWM design; a daughter needing more devices decodes
locally. |

```

2.4 LED interface Emitters are the board's real load and their mounting is **not settled** - on-board emitters versus an off-board module on a separate heatsink with an internal wiring harness is question 5, and it interacts with ICD s9's external-connector ban (an internal harness inside the enclosure is not an external connector, but this needs confirming, not assuming). Optics (PAR-REQ-14 wide wash beam from a 2.5 m ceiling, PAR-REQ-15 diffusion sufficient that R/G/B/W mix before reaching a surface) may or may not be in this run's scope - question 8.

2.5 Status / test points No requirement stated. **ASSUMED:** no user-visible indicators on this board - CAR-REQ-09 status indication lives on the carrier. Any test point fitted must carry the ICD s9 bench-hazard warning: the entire fixture floats at PoE potential, and an earthed scope probe or non-isolated USB-UART adapter does not merely risk damage, it **breaks PD signature detection outright** (detection currents are a few hundred microamps).

3. Power

3.1 The envelope (binding)

```

| | 802.3af (first build) | 802.3at (upgrade path) |
|---|---|---|
| **Sustained power to this daughter, all rails combined** | **8.6-9.3 W** |
**18.7-20.0 W** |

```

D-01 is closed: the carrier's PD front end, converter, magnetics and thermal design are sized for **Type 2 (at)**, but classification is programmed to **Type 1 (af)** for the first build via a class resistor. The upgrade is a resistor change plus a PoE+ switch, **no respin**. D-01 says explicitly: **do not allow any other component to become the part that pins the design to Type 1.** On this board that clause lands on the LED drivers, the emitters and the heatsink - see question 2, which is the highest-consequence open item in this document.

3.2 Per-rail ceilings (ICD s6.2) - individual ceilings, the TOTAL binds

```

| Rail | Sustained af | Sustained at | Peak (ms, local bulk) | Hardware fault
ceiling |
|---|---|---|---|
| '+48V_SW' | 0.25 A (12.0 W) | 0.50 A (24.0 W) | 1.0 A | 1.0 A eFuse, **latch
off** |
| '+12V' | 0.75 A (9.0 W) | 1.25 A (15.0 W) | 2.0 A | 2.0 A converter OCP |
| '+3V3' | 0.25 A (0.83 W) | 0.25 A | 0.50 A | 1.0 A converter, limit >= 1.3 A |

```

****Derived, and load-bearing:** the +12V rail alone cannot cover this board's envelope at either class.** At af it needs 0.72-0.78 A against a 0.75 A ceiling (right at the edge); at at it needs 1.56-1.67 A against a **1.25 A** ceiling, which it misses by 3.7-5.0 W. ICD s6.3 is unambiguous: **anything*

above 1.25 A at the at operating point must be taken on **+48V_SW***, and the 1.25 A ceiling is a thermal limit on the carrier's 48->12 converter in a sealed box, not a current rating that can be argued with. Taking power on **+48V_SW** instead of **+12V** is also worth **0.67 W (af) / 1.30 W (at)** of extra delivered power because it skips a conversion.

ASSUMED: for an **af-sized** board, LED power is taken from **+12V**, which is D-02's stated purpose for that rail (so 6-8 fixtures do not each duplicate a ≥ 60 V converter). If question 2 answers "size for at", that assumption is void and P1 must take at least part of the load from **+48V_SW**, inheriting the bleed path, the 0.60 mm creepage regime, 100 V capacitors, 0805-or-larger resistors across the 48 V domain, and the inrush obligation below.

3.3 Hard rules inherited from the ICD

- **Hardware backstop for PAR-REQ-11.** The clamp is a firmware rule, but firmware fails. With **ENABLE** asserted and every PWM stuck at 100 % (hung MCU; the 2 s watchdog fade cannot help if the MCU is the thing that hung), the board's total draw must remain ****below the carrier's hardware fault ceilings**** - 12 V OCP 2.0 A, 48 V eFuse 1.0 A latch-off, PSE overload timer ~ 50 -75 ms. A board that can only meet the budget when firmware is healthy will brown out or latch off the whole fixture. This sizes the per-channel drive current and is not optional.
- **'+48V_SW' is dead at power-up and stays dead for hundreds of ms.** The carrier's load switch is a compliance part (802.3 caps PD port capacitance at ~ 180 uF); it closes only after firmware asserts **ENABLE**. Design for it.
- **No first-mate/last-mate sequencing exists.** The board must tolerate 48 V arriving before or after 3.3 V, in either order.
- ****No path may energise anything on this board from +12V or +3V3 while +48V_SW is off**, or the 802.3 inrush compliance is defeated behind the switch's back.**
- **Inrush is this board's responsibility (CAR-REQ-14).** Size the limiter against the **PD's 1.0 A operating current limit**, not the connector's 5.4 A rating; sizing against the connector is the classic way to trip the PD's 800 us foldback deglitch and brown out the fixture.
- **If any 48 V net is tapped: a bleed path is mandatory (CAR-REQ-17 / STR-REQ-10).** The carrier deliberately fits no series diode on **+48V_SW**, so the daughter's bleed is not stranded above the carrier's 100 k.
- ****0.60 mm outer-layer copper clearance around every 48 V net, board-wide, is NOT inherited automatically**** - this board's DRC must be set up for it at P5 (IPC-2221B B2, 51-100 V band, 57 V worst case). The 0.13 mm coated column is not claimable: LPI soldermask is not a qualified conformal coating and **check_creepage.py** implements only the uncoated columns, so a layout designed to 0.13 mm fails P8 with no waiver mechanism.

3.4 LED-side budget (ESTIMATES, clearly marked as such) GUESS, for question sharpening only, to be replaced by a real power table at the P1/P2 design gate (03 review gate 1):

- Daughter housekeeping (driver quiescent, ID/EEPROM, sense): **~ 0.3 -0.5 W**.
- Switching CC driver efficiency: **~ 88 -92 %**.
- Therefore electrical power reaching the emitters: **~ 7.6 -8.5 W (af) / ~ 16.5 -18.4 W (at)**.
- A typical 4-in-1 RGBW multi-die emitter (R AlInGaP ~ 2.1 -2.9 V, G/B InGaN ~ 3.0 -3.6 V, W ~ 2.9 -3.4 V) draws roughly **4.0-4.7 W at 350 mA/die** and **8.4-9.8 W at 700 mA/die** with all four dies on. So **one** such emitter at ~ 600 -700 mA/die consumes the entire af envelope by itself; the at envelope buys roughly double.

- Note for P1, not a decision here: PAR-REQ-10 forbids burning the R-vs-GBW forward-voltage spread in a shared linear element, and a 12 V rail against a 2.1-3.6 V single die is a 3.3-5.7x step-down. That combination rules out shared linear current control and points at either per-channel switching regulators or multi-die series strings per channel. It also interacts with question 3 (package choice).

No battery, no charging, no mains anywhere in this system.

4. Environment

- Deployment: indoor, basement/garage room ~5 m x 7 m x 2.5 m, 8-12 fixtures, ceiling-mounted at ~2.5 m. **ASSUMED:** ceiling mount, beam pointing into the room; no vibration or shock requirement; no ingress rating (question 6 decides sealed vs vented, which is a thermal question, not an IP one).
 - **ASSUMED:** external ambient 0-40 C. Not stated anywhere in the briefs.
 - **Internal enclosure air reaches 56 C (af) / 69 C (at)** per ICD s7.6. This is the ambient the emitters and drivers actually see, not room temperature.
 - **Continuous duty with no relief.** Unlike the strobe, this board runs hot all the time (03 review gate 4, and H1-Q5 restates it). Every thermal number must be a steady-state number.
 - Heat this board must reject (ESTIMATE: LEDs convert ~20-35 % of electrical input to light, the rest to heat): **~6.5-8.4 W (af) / ~14-17 W (at)**, plus the carrier's own 2.4 W (af) / 3.7 W (at) in the same box.
 - **This is the riskiest number in the document.** With internal air at 69 C and a red-die junction that wants to stay near 85-100 C for colour and lifetime stability, the junction-to-internal-air thermal budget at the at operating point is roughly **1.0-1.1 C/W**, and at af roughly 2.3 C/W. A sealed plastic box has no convective path out. Questions 6 and 7 exist because the at case very likely does not close without either ventilation or an external heat path, and H1-Q5 constrains both (non-conductive enclosure, heatsink NOT user-accessible).
 - Connector contacts are rated -40 to +105 C, so they are not the limit.
-

5. Size & mounting

All of the following is the **common LUMINA footprint** from ICD s7.1, which every board inherits. It is proposed, not yet confirmed with the human - MECH-02 requires the outline to be closed at H1 **before P5**, because `board_init.py` binds the outline permanently and ai-ee has no outline-shrink step. See question 12.

```
| Item | Value |
|---|---|
| Outline | **100.0 x 80.0 mm** |
| Corner radius | **3.0 mm**, all four corners |
| Mounting holes | **4x M3 (3.2 mm) at 5 mm inset** (a 90 x 70 mm rectangle)
**plus a 5th M3 at (46, 74)** |
| Thickness | 1.6 mm |
| Coordinate origin | board top-left, x right, y down |
| Mated board-to-board height | **11.0 mm** hard-seated |
| Standoffs | 5x M3 female-female, 11.0 mm |
```

5.1 The RJ45 notch - hard requirement from P2 onward A 30 x 26 mm notch in the TOP edge, region (6, 0) - (36, 26). The carrier's board-edge magjack is ~15 mm tall against an 11.0 mm stack, so it protrudes ~4 mm above this board's underside. The outline rectangle, corner radius and 5-hole pattern are unchanged; only this local relief differs. It is also the **primary anti-180-degree interlock** (CAR-REQ-16): rotated, the notch lands at the bottom edge and the board presents solid material over the jack, so the boards cannot be forced flat. That is a mechanical stop, not a warning.

5.2 Exclusion zones (ICD s7.6) - all four apply to this board

Zone	Region	Rule
---	---	---
RJ45 relief	(6, 0) - (36, 26)	cut away entirely (s5.1)
DC-DC hot zone	(2, 46) - (36, 68)	**no LED drivers and no aluminium electrolytics** - the carrier's 48->12 converter dissipates up to 1.25 W directly below, and in a stacked mezzanine this board's parts sit vertically over it
Antenna column	(88, 25) - (100, 55)	**no copper on any layer, no metal component** - the carrier's ESP32-S3 PCB antenna is 11 mm below and Wi-Fi is a supported control path (H1-Q8)
Recovery header	(76, 0) - (98, 20)	keep clear enough for a 6-way jumper lead with this board fitted, or accept board removal for firmware recovery

5.3 Connector positions - PROVISIONAL, and this board is blocked on them ICD s7.2 gives J3 body (14, 68)-(34, 78) with position 1 at (15.3, 69.3), H5 at (46, 74), J4 body (56, 68)-(88, 78) with position 1 at (57.3, 69.3). ****ICD s7.2 is the one section not frozen at H1****: the carrier owner must compare placed positions after its own P6, correct them, and re-issue. Daughters are explicitly blocked on that re-issue. See question 16.

Mating geometry to verify at P4/P6, not to re-derive: this board's sockets are **bottom-side, facing down**, and mate with a top-side header on the carrier. Plan-view (x, y) coordinates are shared between the two boards, but pin numbering mirrors across the row on a bottom-side footprint - pin 1 alignment must be checked in the mated view, not assumed from the footprint.

5.4 Pipeline facts that constrain P5 (verified in this repo, not assumed)

- `board_init.py --mounting-holes N` places holes at **inset = margin / 2** (confirmed in the script). The ICD's 5 mm inset therefore requires **'-margin 10'**, not the default 6. The corner radius is separately clamped in the worker; read the report's `corner_radius` field and `worker_notes` rather than assuming the requested 3.0 mm was honoured.
- **'board_init.py' has NO notch/cutout option** (`--outline` takes only `auto` or `WxH`), and `kc.py` exposes no outline-editing subcommand. The mandatory 30 x 26 mm relief must therefore be produced by a direct `Edge.Cuts` edit of the `.kicad_pcb` after `board_init`, with every downstream consumer (`place_*`, `planes_gen`, `dfm_check`, `fab_export`) re-verified against the modified outline. **Flagged as an implementation risk for P5**, not a requirements question - the notch itself is non-negotiable.
- Placement keepouts ARE supported: `constraints.json` accepts `placement.keepouts: [{"rect": [x1,y1,x2,y2], "reason": ...}]` and the annealer treats them as fixed obstacles. All four ICD s7.6 zones plus the notch region must be declared there.

5.5 Height Board-to-board is 11.0 mm. **Total height above this board is unknown** and depends on the emitter/heatsink/optic stack (questions 5-8). Not stated anywhere in the briefs.

6. Quantity & budget

- **Quantity: 6-8** par fixtures out of 8-12 total (03 s1). Exact count and spares not stated - question 14.
- System budget: **\$500-1000 for fixtures + switch + enclosures** (00 s1). ICD s1.1 rejects a Samtec pair at \$15.30 "over half the \$30/board target", so a **\$30/board** figure exists in the carrier run - but no unit cost target has been stated for *this* board. With 8-12 fixtures, a managed PoE+ switch (note: D-01's upgrade path implies PoE+, which is materially more expensive than the PoE switch the original budget assumed), enclosures, carriers, strobes and pars all inside \$500-1000, this is tight. Question 14.

7. Assembly

- **ASSUMED:** JLCPCB fabrication and PCBA, consistent with every other board in this repo. Not stated in the LUMINA briefs.
- ****Both ICD connectors are 2.54 mm THT sockets that must be reverse-mounted on the bottom side.**** That is not a normal JLC PCBA process, so the realistic routes are hand-soldering them post-PCBA or finding a bottom-side SMD equivalent on the same land pattern (ICD s7.3 permits "a bottom-side SMD equivalent"). Question 13.
- High-power emitters, if on-board, need a large thermal pad and a via farm on 1.6 mm FR4 - or an MCPCB, which the common footprint does not provide. Interacts with questions 5 and 13.
- Sidedness: bottom side already carries the two sockets, so the board is double-sided by construction. Question 13 confirms the process.

8. Compliance / safety flags

```
| Flag | Status |
|---|---|
| Mains voltage | **No.** PoE only. |
| Batteries | **No.** No battery, no charging anywhere in the system. |
| Motors | **No.** |
| **> 30 V** | **YES, unconditionally.** '+48V_SW' is present on J3 whether or
not this board taps it. PD rail 37-57 V nominal, **57 V worst case** (IEEE
802.3 PSE maximum). Consequences: 0.60 mm outer-layer clearance around every
48 V net board-wide; 100 V capacitors (63 V is not enough at 57 V once ceramic
DC-bias derating applies); any resistor across the 48 V domain must be 0805 or
larger, or split in series; the clearance applies through the board (inner 0.10
mm / outer 0.60 mm under any 48 V antipad); mandatory bleed path if tapped.
**Insulation class is functional only** - 57 V is below the IEC 62368-1 ES1
limit of 60 V DC, so no personnel safeguard and no safety-mandated creepage
applies, and an unearthed PD needs **no MOV-to-earth surge network** (do not
copy one from a reference design). |
| High current (> 3 A) | **Conditional - architecture must report and
re-flag.** No connector rail exceeds 1.25 A. But LED-side per-channel current
is unbounded by the ICD: a single low-Vf red channel sized near the whole
envelope (e.g. 8 W into a ~2.4 V string) would be ~3.3 A. P1/P2 must publish
per-channel drive currents; if any exceeds 3 A this flag becomes unconditional.
|
| RF transmit | **No radio on this board** - but the carrier's ESP32-S3 radio is
a *supported control path* (H1-Q8), so the (88, 25)-(100, 55) antenna column is
```


a hard keepout (no copper on any layer, no metal component) and switching LED drivers 11 mm above a 2.4 GHz PCB antenna are a desense risk that layout sign-off must address. |

| ****Floating PoE potential**** | ****Applies to everything this board touches.**** The entire fixture is non-isolated and floats at PoE potential: this board, its drivers, its LED wiring, and any heatsink the LED module sits on. 802.3 compliance is achieved *only* by there being no accessible external conductor. This is ICD s9 / 'decisions.md' OPEN-C and it is ****unresolved**** - questions 5 and 7. |

| ****Continuous hot surface**** | High-power LEDs at 8.6-20 W continuous inside a non-conductive enclosure whose internal air already reaches 56-69 C. Enclosure touch temperature and plastic deformation are real; H1-Q5 requires the heatsink to be non-user-accessible. |

| ****Photobiological (IEC 62471)**** | A high-power RGBW source with a wash optic is plausibly Risk Group 1-2 at close range (blue-light hazard). Not a certification requirement for a private installation, but worth a note in the design doc rather than discovering it by looking into a fixture. |

| Certification | 'ASSUMED:' private installation, own use - no CE/FCC/UL conformity assessment path required. |

Nothing in section 8 is guessed. The >30 V condition and the floating-PoE condition are stated facts from the ICD; the >3 A condition is explicitly left open for the architecture to answer.

9. Open questions

Questions 1-4 are the four the orchestrator already queued, sharpened with the numbers above. Questions 5-16 are everything else this design genuinely needs. Each offers a recommendation. Questions **2, 6, 7 and 12** are the ones that change the board most, and 12 and 16 must be answered before P5 regardless.

1. Total-power clamp policy (PAR-REQ-11). Four channels cannot all run at 100 % - the sum of four channels at full drive exceeds the 8.6-9.3 W (af) envelope by roughly 30-60 % on any sane sizing. When the host asks for more than the fixture can deliver, what should the fixture protect? (a) **Preserve hue**: scale all four channels by the same factor. The colour is exactly right; the fixture is simply dimmer. White (all four on) is the worst case and gets dimmest. (b) **Preserve brightness**: hold the total at the cap and reallocate between channels. Brightness is stable; the hue drifts toward whichever channel is cheapest in watts, and drifts *differently* per fixture, which is exactly the artifact PAR-REQ-06 exists to prevent. (c) Hybrid: preserve hue, let the white channel absorb the deficit - this desaturates at high intensity, the "visible artifact, not graceful degradation" the brief warns about. **RECOMMEND (a)**. Uniform hue-preserving scaling. Colour consistency across 6-8 fixtures washing one wall is this board's reason to exist; a uniform brightness loss is invisible, a hue split across fixtures is not. Note that whichever answer is chosen, the **hardware backstop in s3.3 is still required**

- the board must be electrically incapable of exceeding the carrier's fault

ceilings with every PWM stuck at 100 %.

2. Size the LED engine for af or at? D-01 kept the *carrier* at-capable and af-classified, and told us not to let any component pin the design to Type 1. Does that clause bind this board's LED engine too? (a) **Size for af (8.6-9.3 W)**: smaller emitter, smaller heatsink, cheaper, thermally plausible in a plastic box - but a PoE+ upgrade then buys nothing on the par because the emitters and heatsink cannot use the extra ~10 W, and getting it later is a respin of this board. (b) **Size for at (18.7-20.0 W), run clamped at af**: the upgrade is a firmware constant. But it roughly doubles heatsink area and emitter cost per fixture, and - derived above - ****it forces at least 3.7-5.0 W to be taken from +48V_SW**** because the +12V rail tops out at 15.0 W, which means a ≥ 60 V input stage on this board after all, the thing D-02's 12 V rail was created to avoid on 6-8 fixtures.

****RECOMMEND (a)** for the first build, with a written note in the design doc that the par does not follow the carrier's at upgrade.** The par is a sustained wash, not the fixture that needs the headroom; the strobe is. Spending the at budget on the par costs money and heatsink on all 6-8 units to buy brightness the profiles ask of the strobe. But this is a lighting-design call, not an engineering one - if a 9 W par reads as underwhelming, (a) is a respin.

3. LED package: four discrete emitters or an integrated RGBW multi-die? (a) **Integrated 4-in-1** (single package, four dies on one thermal slug): mixes better optically and largely solves PAR-REQ-15 fringing; one thermal interface; one binning transaction. All four dies share one junction temperature, so the hottest channel heats the others, and red (AlInGaP) is both the most thermally sensitive and the one that gets cooked. (b) **Four discrete emitters**: independent thermal management, cheaper and easier sourcing, per-channel binning - but four separate optical sources need real diffusion to avoid visible R/G/B/W shadow fringing on a wall (PAR-REQ-15), and four packages spread over the board is four thermal paths to solve. **RECOMMEND (a) integrated 4-in-1**, on the grounds that PAR-REQ-15 is a hard optical requirement and mixing is the one thing an integrated package gives for free. Note the interaction with s3.4: a 12 V rail against a single 2.1-3.6 V die is a 3.3-5.7x step-down, which pushes toward per-channel switching drivers either way.

4. A fifth channel (amber)? PWM budget is free - the connector carries 8 channels and the par uses 4, and PWM4 sits on LEDC timer 1 which can be programmed to the same 13-bit/9.766 kHz. So the cost is not connector budget, it is: one more driver, and a fifth channel drawing from the **same** 8.6-9.3 W envelope, making the question-1 clamp bite harder in every mixed colour. P3 French melodic leans on gold/amber and RGBW amber is a mix rather than a true emitter. Caveat worth weighing: amber AlInGaP loses output at roughly -0.5 to -1 %/C of junction temperature, roughly twice as badly as the InGaN channels, in a box whose internal air is already at 56-69 C continuously - so a calibrated amber will drift visibly as the fixture warms up unless it is temperature-compensated in firmware. **RECOMMEND: no fifth channel on rev A.** Add it only if question 2 answers "size for at" (where there is power to spare). At af it dilutes the envelope across five channels and adds the most thermally unstable emitter in the catalogue to a board that already has a thermal problem.

****5. LED mounting and wiring: on-board emitters, or an off-board LED module on a separate heatsink?**** (a) **On-board**: emitters on this PCB's top side, heat through thermal vias into a heatsink bolted to the board. No wiring at all, so ICD s9 is trivially satisfied - but 1.6 mm FR4 is a poor thermal path, the common footprint gives no MCPCB, and the emitters' position is then dictated by the connector/keepout geometry rather than by the optics. (b) **Off-board module** on its own MCPCB/heatsink, connected by an internal wiring harness (5-9 conductors). Best thermal and optical freedom. ****ICD s9 bans a daughter from adding "an external connector of any kind"** - a harness that never leaves the enclosure is not an external connector on any reasonable reading, but this needs confirming, not assuming**, because if it is read strictly then option (b) is illegal and option (a) is the only route. ****RECOMMEND (b)**, with explicit confirmation that an internal wire-to-board connector (e.g. a JST-family header) is permitted** on the grounds that ICD s9's intent is "nothing penetrates the enclosure wall", and s9 itself already anticipates the off-board case ("if the LED module is on a separate heatsink (Q4a default), that heatsink and its wiring are at PoE potential too"). If the answer is a strict no, please say so plainly - it forces (a) and changes the thermal design completely.

6. Is the enclosure sealed, or vented? ICD s7.6 says "sealed box" with internal air at 56 C (af) / 69 C (at); H1-Q5 says plastic, non-conductive, with the heatsink not user-accessible. No ingress requirement is stated anywhere. This matters more than any other mechanical question: from s4, a *sealed* plastic box gives a junction-to-internal-air budget of ~2.3 C/W at af and ~1.0 C/W at at, and the at case very likely does not close. (a) Fully sealed (no openings at all). (b) **Vented** - louvres or a labyrinth sized so no finger or 4 mm probe reaches live parts or the heatsink, preserving H1-Q5's non-accessibility while allowing convection. (c) Sealed but with the heatsink forming part of the enclosure wall (see question 7). **RECOMMEND (b) vented with a finger-guard geometry.** It is an indoor basement/garage installation with no stated dust or moisture

requirement, and convection is close to free. Sealing costs real money in heatsink area and caps the fixture's output.

7. Is an external or enclosure-integrated heat path allowed? ICD s9 and `decisions.md` OPEN-C leave this open and it is the sharp end of the non-isolated topology: any heatsink bonded to this board's ground ****floats at PoE potential****, and "if the heatsink is touchable, metal, or shares a mount with anything earthed, the non-isolated topology is non-conformant." (a) Heatsink entirely internal, no metal reaches the outside. Thermally worst, compliance-simplest. (b) Heatsink forms part of the enclosure but is shrouded behind a plastic guard - not touchable, and not sharing a mount with anything earthed. (c) Exposed metal heatsink - **would break the compliance argument** unless the whole fixture is re-isolated, which is out of scope. **RECOMMEND (b)**, plus a hard rule that the ceiling mount is non-conductive and bonded to nothing. Also please confirm: should the LED module's metal substrate be electrically tied to board GND, or thermally coupled but electrically floating? (Recommend: floating from board GND, on a dielectric, so a single insulation failure does not put 48 V-referenced potential on the heatsink.)

8. Does this run own the optics? PAR-REQ-14 (wide wash beam from 2.5 m, overlapping coverage from 6-8 fixtures) and PAR-REQ-15 (diffusion sufficient that the colours mix before reaching a surface) are stated as requirements on this board, but a PCB run can only choose the emitter and provide mechanical provision for a lens or diffuser. (a) This run selects the emitter and provides lens-holder mounting provision; the diffuser/lens is an enclosure item chosen later. (b) This run owns the full optical chain including the lens/diffuser part numbers and beam-angle verification. **RECOMMEND (a)** - a PCB pipeline cannot verify a beam angle, and PAR-REQ-14 / 15 should be carried as enclosure requirements in the design doc so they are not silently lost.

9. What does "5-10 % of full output" mean in PAR-REQ-01? This changes the required driver PWM bandwidth by roughly 40x and it is the single most likely reason this board fails its own review gate 2. At 13-bit / 9.766 kHz the period is 102.4 us and one LSB is **12.5 ns** - no constant-current driver on the market resolves that, so the real dimming floor is the driver's PWM settling time, not the PWM word length. (a) **5-10 % of PWM duty**: on-time 5.1-10.2 us. Most switching CC drivers manage this at 10 kHz with some amplitude error. (b) **5-10 % of perceived brightness** (i.e. after gamma ~ 2.2): duty is 0.137-0.63 %, on-time **141-646 ns**, which is at or below the settling time of essentially every CC driver and is where "visible stair-stepping during slow fades" is actually born. **CORRECTED 2026-07-28 (orchestrator, verified)**: this line previously read "on-time 1.4-6.1 us", a 10x error. The duties were right; the on-times were not. $0.05^{2.2} \times 102.4 \text{ us} = 140.6 \text{ ns}$ and $0.10^{2.2} \times 102.4 \text{ us} = 646.1 \text{ ns}$. The correction is load-bearing: it moves (b) from "several drivers manage it" to "no surveyed CC driver manages it with PWM alone" - the best part found in P1 (TPS92515HV) specs a 0.2 % minimum duty at 10 kHz, i.e. a 200 ns floor, which is above the 141 ns that 5 % perceived requires. See `research/led-driver.md` and `research/spec-dimming.md`. **RECOMMEND: assume (b)**, because that is what a lighting person means by "dim to 5 %", and it is the harder requirement - designing for (b) satisfies (a) automatically. Please confirm, because it materially narrows the driver shortlist and is the requirement PAR-REQ-09 is really about.

10. Calibration path (PAR-REQ-17) - and who does the measuring? The premise that this is EEPROM *versus* the resistor divider is **not what the ICD says**. ICD s2 and s3.3: the carrier fits the top leg of the ID_ADC divider and the daughter fits the bottom leg **and** an I2C EEPROM may ride the shared I2C bus; a second dedicated ID pin was deleted precisely because the EEPROM does not need one. So the two mechanisms coexist: ****the divider says what board type this is, the EEPROM stores per-unit calibration.**** A 24C02/24C32-class part is a few cents and one address allocation (this board owns the whole I2C address space - the carrier reserves nothing - so EEPROM and any digital temp sensor must be allocated non-colliding addresses). **RECOMMEND: fit both**. The real open question is downstream: PAR-REQ-06 requires measured per-channel scaling, so ****what instrument measures 6-8 fixtures?**** (colorimeter / spectrometer / phone-camera comparison / none). If the answer is "nothing", the EEPROM ships empty and colour matching falls back entirely on binning (question 11), which changes the sourcing strategy.

11. LED binning versus the assembly route. PAR-REQ-16 requires binning specified in

the BOM for all four channels; "unbinned parts across 8 fixtures will not satisfy PAR-REQ-06". But JLC's catalogue does not generally let you specify a chromaticity/flux bin, so binned high-power emitters usually mean hand-sourcing from a distributor and hand-placing them. (a) Hand-source binned emitters and place them by hand (or as a separate LED sub-assembly), everything else JLC PCBA. (b) Accept whatever bin JLC ships and rely entirely on per-fixture calibration (question 10) to match the fixtures. (c) Both: loose binning plus calibration. **RECOMMEND (c)** - buy the tightest bin available from a distributor for one batch of 6-8 fixtures (they will then be same-reel and closely matched anyway) **and** fit the calibration EEPROM. Note (b) alone is only viable if question 10 has a real measurement instrument behind it.

12. Confirm the board outline - MECH-02, needed before P5. The ICD proposes the common footprint: **100.0 x 80.0 mm, 3.0 mm corner radius, 1.6 mm thick, 4x M3 at 5 mm inset plus a 5th M3 at (46, 74), and the mandatory 30 x 26 mm notch at (6, 0)-(36, 26). **ai-ee has** no outline-shrink step** - whatever is passed to `board_init` at P5 is permanent, so this must be confirmed, not inherited silently. **RECOMMEND: confirm 100.0 x 80.0 mm as-is.** The carrier and daughters share one enclosure and mate through the connector, so the outlines are not independent, and after the four exclusion zones and the notch this board has roughly 60 x 74 mm of genuinely usable area for drivers plus whatever the LED scheme needs. Please confirm explicitly, including whether the enclosure has any dimension this footprint must fit inside that has not been stated.

13. Assembly method and process. Not stated anywhere in the LUMINA briefs. (a) JLCPCB fabrication + PCBA, JLC Basic parts preferred where they exist (note: the ICD connectors are Extended, unavoidably - there is no Basic board-to-board part). (b) Hand assembly. Two sub-questions that need answering with it: **are the two 2.54 mm sockets hand-soldered post-PCBA** (they are THT and must be reverse-mounted on the bottom side, which is not a normal JLC process), and **are high-power emitters hand-placed** (question 11)? ****RECOMMEND: JLC PCBA for everything it can do, with the two sockets and the emitters hand-soldered afterward.**** 6-8 boards is a quantity where two hand operations per board is a couple of hours total, and it removes both awkward processes from the critical path.

14. Build quantity and unit cost target for THIS board. The brief says 6-8 pars of 8-12 fixtures; no per-board cost target is stated (the \$30/board figure in ICD s1.1 belongs to the carrier). The \$500-1000 system budget also predates D-01's at-capable power stage, and the upgrade path implies a **PoE+** switch, which is materially more expensive than the PoE switch that budget assumed. ****RECOMMEND: confirm 8 boards built (6-8 deployed plus spares) and set an explicit BOM target for this board.**** A defensible starting number is **\$25-35/board** excluding the LED module and heatsink, with the emitter and heatsink budgeted separately because question 2 can double them.

15. 'ID_ADC' code allocation. ICD s3.3: "Board-type codes are allocated by the carrier owner, not chosen by daughters." This board needs its allocated bottom-leg resistor value (and the resulting ADC code) before P4 can place the part. **RECOMMEND: request the LUM-PAR-A code from the carrier owner now;** if none exists yet, ask for one to be allocated rather than picking a value here - picking one is exactly the silent divergence ICD-01's preamble forbids.

16. Sequencing against ICD s7.2 (the one un-frozen section). The J3/J4 coordinates in s7.2 are provisional until the carrier owner confirms them after its own P6 and re-issues the ICD, and daughters are explicitly blocked on that. Everything else in the ICD is frozen. (a) Wait for the re-issued ICD before starting this run. (b) ****Proceed through P1-P4 (architecture and schematic, which do not depend on mm coordinates) and hold at P5**, where the outline and connector placement are bound permanently. **RECOMMEND (b),** with the s7.2 coordinates treated as unconfirmed everywhere they appear, and an explicit re-check against the re-issued ICD before `board_init` is run. Please confirm this is acceptable, and if the carrier's P6 has already completed, point this run at the re-issued ICD instead of the DRAFT-A copy in `brief/`.**

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	D-01/D-02/D-03 inherited as closed; par power budget 8.6-9.3 W af / 18.7-20.0 W at	05-lumina-closed-decisions.md is binding and supersedes the ~8.5 W figure in briefs 00/03 (derived from Si3402-B, a part this system does not use)
P0	ICD-01 (brief/06-connector-icd.md) treated as a frozen hard input; no local variant permitted	ICD s10 change process: a daughter that cannot meet it raises a blocking issue against LUM-CAR-A, it does not implement a variant
P0	Common footprint adopted from ICD s7.1: 100.0 x 80.0 mm, R3.0 corners, 5x M3 (4 corners at 5 mm inset + H5 at (46,74)), 1.6 mm FR4, with a mandatory 30x26 mm RJ45 notch at (6,0)-(36,26)	H1-Q4 closed the stack at 11.0 mm; the carrier's board-edge RJ45 is ~15 mm tall so the notch is a hard mechanical stop and the anti-180-degree interlock (ICD s7.4)
P2	Re-baselined onto ICD rev A2 (copied from boards/lumina-carrier/architecture/connector-icd.md)	Two changes bind this board: (1) board-wide HV clearance raised 0.60 -> 0.635 mm per TPS2378 vendor layout guidance, daughters inherit it; (2) new s8.4 - the carrier's TPS16630 load switch has a mandatory 10k SHDN pull-down, so a daughter may assume an unprogrammed/crashed/reset/brownout carrier presents 0 V at J3, not 48 V. Rail limits s6.2 and mechanical s7.1/s7.2 are UNCHANGED.
P2	LED stage stays on +12V (baseline); the P1 '48V is 16x better for dimming slew' argument is WITHDRAWN as an artifact of holding L fixed	Verified algebraically by the orchestrator: with L sized for the same ripple fraction k, $t_{rise}+t_{fall} = 1/(f*k)$, independent of Vin. At 700 kHz / k=0.30 both rails give 4.762 us (48V only redistributes 3.81/0.95 us between rise and fall). That is 33.8x the 141 ns PAR-REQ-01 budget, so NO rail or inductor choice reaches the requirement by gating the converter - it needs shunt-FET dimming either way. The 48V branch therefore buys nothing on fidelity and costs 0.635 mm creepage, 100 V caps, bleed path and daughter-owned inrush.
P2	Stackup JLC04161H-3313, 4 layers, 100.0 x 80.0 mm, 1.6 mm	Switch-node containment 11 mm above a live 2.4 GHz antenna (H1-Q8 made Wi-Fi a supported control path), continuous GND reference under jitter-sensitive PWM, B.Cu consumed by two reverse-mounted sockets, ~120 parts in ~45 cm2. No controlled-impedance net exists on this board.
P2	Dimming topology: shunt FET per channel across the LED string	Closes the PAR-REQ-01 TIMING wall: TPS92515HV direct PWM floors at 200 ns vs a 141 ns requirement, but its shunt mode is spec'd 10,000:1 = 10.2 ns min pulse. Implemented T ~ 50-100 ns, 1.4-2.8x margin. The RESOLUTION wall (17 bits needed, LEDC caps at 14) is NOT closable on this board and needs carrier firmware dithering (CR-5).

Phase	What	Why
P2	Design point 150 mA/die, 4 emitter packages, 2S2P per channel; emitters off-board on aluminium MCPCB	Full white = 0.718 A on +12V, 4 % under the sustained ceiling and 2.8x under OCP, so the board is electrically incapable of exceeding its budget with every PWM stuck at 100 %. Off-board MCPCB because 1.6 mm FR4 misses the junction budget by 2-3x. >3 A conditional flag CLOSED (max 0.30 A/channel).
P2	No fifth (amber) channel on rev A; fit BOTH ID mechanisms (ID_ADC bottom-leg divider + I2C EEPROM)	Amber dilutes an af envelope that full white already fills to 96 % of the rail ceiling. ICD s3.3 makes the divider and the EEPROM coexist rather than alternatives - the divider gives board type, the EEPROM per-unit calibration - so PAR-REQ-17 is met without spending a pin.
P2	Q1 CLOSED: enclosure sealed with LED heat conducted through the wall (not vented)	Owner decision: keeps dust out of a basement/garage install over years, and the strobe run independently recommended the same, so both fixtures share one enclosure design. 3.21 W internal -> 38 C air. The wall-conduction path becomes load-bearing for the thermal case and must be written as an explicit testable ENC-x criterion, not an assumption.
P2	Q2 CLOSED: emitter is RGB 3-in-1 + separate white discrete, superseding D3's integrated 4-in-1	Owner decision, taken knowingly against this run's recommendation. Deciding factor: the thermal case now rests on a published 125 C Tj max and 3.5x stock depth rather than on an ASSUMED 12 K/W Rth from a sole-source vendor publishing neither Rth nor Tj max. Retires the 20 K/W marginality risk this run flagged. Consequences owned here: PAR-REQ-15 white/colour fringing becomes the live risk; the entire sustained-power table is stale and must be re-derived; 145 mA/die is reopened.
P2	Q3 CLOSED: PAR-REQ-01 met by hardware + firmware. Strict reading (5% perceived, gamma 2.2) binds.	Carrier firmware commits to PWM-domain dithering >=3-4 bits (CR-5 accepted). Shunt-FET hardware half stands as designed (T ~50-100 ns vs 141 ns). FIRMWARE DEPENDENCY must be named in the design doc so it cannot be silently dropped: without the dither the requirement fails by 4.4-8.9x and no hardware on this board can fix it. Dither must be PWM-domain, not 60 fps frame-domain - a 4.4% dither at 60 Hz breaches IEEE 1789 NOEL on its own.
P2	Q4 CLOSED: same-reel sourcing; 24C32 EEPROM fitted but shipped empty; PAR-REQ-16 and PAR-REQ-17 to be formally amended with original text quoted	No measurement instrument exists today, so colour matching rests on same-reel binning. The EEPROM is fitted so that decision stays reversible. Amending rather than quietly failing PAR-REQ-16 keeps the record honest about what was changed by decision.

Phase	What	Why
P2	PAR-REQ-16 and PAR-REQ-17 formally amended (AMD-01, AMD-02 in requirements.md s0), original text quoted	AMD-01: a chromaticity/flux bin is not an orderable attribute on LCSC/JLC and the selected vendor publishes none, so the BOM line was unbuildable; replaced by single-reel procurement for the whole build. This is a REDUCTION in guarantee, not an equivalent - same-reel is an empirical correlation, a published bin is a contractual limit - so PAR-REQ-06 is carried at reduced confidence with bench verification. AMD-02: hardware (divider + 24C32) is fitted; the calibration DATA is deferred because no colorimeter exists for this build.
P2	The RJ45 notch is the coordinator's to implement via a central board_init -cutout flag. This run does NOT hand-edit Edge.Cuts and does NOT shrink the outline.	Both daughters need the notch, so it is being added centrally rather than worked around per-run. The notch stays a hard keepout in constraints.json meanwhile. Coordinator will confirm the flag exists before this run reaches P5. Supersedes the OPEN-7 remediation (hand Edge.Cuts edit) recorded at H1; the OPEN-7 coordinate-translation trap under fixed-outline board_init still stands and must be re-raised at H2.
P3	P5 outline procedure fixed: board_init -outline 100x80 -corner-radius 3 -mounting-holes 4 -cutout 6,0,30,26, then verify geometry against an expected area of 7212.274 mm2	Coordinator added -cutout centrally (commit 18613d3) so both daughters share it. Expected area derived here so P5 has a number to check rather than eyeballing: 100x80 = 8000, minus the 30x26 notch = 780, minus 4 corners at r=3 = $4 \times r^2(1-\pi/4) = 7.7257$. The outline_origin translation is the trap that would otherwise silently misplace every keepout.
P3	Emitter set: 4x XL-HD6070RGB-C-A46L-BD (C22434861) + 4x XL-HD6070UWC-A4-BD (C48586656); 2S2P, 150 mA/die, 300 mA/channel. Headline 0.717 A = 95.6% of the 0.75 A ceiling.	Both live-verified via parts_search this session and spot-checked independently by the orchestrator. The headline survives the package change because BOTH packages publish identical Vf ranges (R 1.8-2.4, G/B/W 2.8-3.4), so the 2S2P/150 mA arrangement falls out the same way - a derived result, not a copy-forward. All nine power rows independently recomputed by the orchestrator and reproduce exactly.
P3	150 mA/die confirmed over 145 (H1-P7 re-derived, not carried forward)	Every genuine HARDWARE ceiling passes at both: 2.8x on the 2.0 A OCP, 3.4x on the TPS2378 0.85 A MINIMUM limit. The only figure 150 fails is the ICD 8.5 W total, which ICD s6.2 itself calls a firmware-governed average, not a trip. Flips to 145 if the carrier owner declares 8.5 W a hardware limit.

Phase	What	Why
P3	Sense resistor 0.75 ohm (not 0.68), nominal 275 mA/channel - the COMPLIANT value is the default	P3 found that 0.68 ohm with IADJ tied to VCC gives a $\pm 5.6\%$ threshold band $\rightarrow$ ILED 284-324 mA $\rightarrow$ 0.775 A worst case = 103% of the ICD 0.75 A +12V sustained ceiling. Orchestrator reproduced it and also costed the proposed IADJ divider: the divider tightens the band but lowers the threshold too, landing at essentially the same light (0.928 vs 0.917 of nominal) for 8 extra resistors - REJECTED. Compliant-by-construction is the right default because the stuck-PWM backstop is exactly the case where firmware is dead and cannot govern the average. Costs 8.3% light vs the H1 point. Reversible: BOM value only, zero layout impact, changeable to P9. Flagged for H2.
P3	Shunt FET switched to TOSHIBA SSM3K2615R,LXGF (C22371361, SOT-23F)	Closes two P3 open items at once. Ciss 150 pF into the 150 R gate resistor is $\tau \sim 22$ ns, so gate edges land in the tens of ns against PAR-REQ-01's 141 ns pulse - the previously selected part was estimated at 100-140 ns, which would have eaten the entire timing budget. And Rds(on) 580 mohm is specified AT 3.3 V, so it is guaranteed on at our gate drive rather than inferred. Also resolves the no-first-tier-brand-in-stock risk. Dissipation 69 mW at 0.345 A peak. Package is SOT-23F not SOT-23 - fp_verify needs extra attention.
P3	Keep the THT reverse-mounted ICD sockets; REJECT the HC-PM254 SMD equivalents	ICD s5.2 derives its 0.84 mm pad-to-pad creepage from the THT land pattern (1.70 mm annulus on a 1.02 mm drill), and that geometry is the carrier owner's frozen number. An SMD land pattern changes it and would require the ICD to be re-issued. Two hand-solder operations across 8 boards is the cheaper side of that trade.
P3	Re-synced ICD copy again; daughters MUST use a 1.70 mm annulus on the connector pads (1.10 mm drill, 0.300 mm/side ring)	The carrier's P3 library work corrected the drill from the assumed 1.02 mm to an actual 1.10 mm, and normalised J3's annulus from 1.80 to 1.70 mm so both frozen connectors share one land pattern. The 0.84 mm pad-to-pad gap and the 1.32x clearance conclusion are unchanged, but the annulus is now an explicit daughter requirement - this is a librarian/fp_verify check-point, not just a note.
P3	0.75 ohm sense resistor ENDORSED by the coordinator; C2934269 recorded as the explicit 0.68 ohm alternate	Coordinator: 'a margin that only holds while software works is not a margin.' Reversion to 0.68 ohm stays live as a one-code swap through P9.

Phase	What	Why
P3	CR-1 CLOSED: R206 = 4.7 kOhm, 1% REQUIRED (not preferred). LUM-PAR-A is ID_ADC code 2, V_ID = 1.055 V.	ICD s3.4 rev A3, NORMATIVE. Verified: $3.3 \times 4700 / (4700 + 10000) = 1.0551$ V. 1% is required because adjacent-code separation is 0.353 V against a ± 0.05 V divider error from 1% parts plus a ± 0.15 V firmware window - a 5% part eats the margin. Cited by section+amendment revision, not the header (the header reads A3 while its own parenthetical lists amendments through A6). The strobe run found its placeholder was set to THIS board's code and would have shipped announcing itself as the par - the value is functional, not cosmetic.
P3	CR-6 CLOSED by ICD s7.7 rev A5 (NORMATIVE): sealed + wall-conducted, internal air 32-33 C at a 25 C room	The carrier re-derived and reconciled. With the LED wall path WORKING only 1.894 W enters box air (6.8-8.1 K rise); with it FAILING 8.5 W enters and gives 56-62 C - which reproduces the rev A2 af figure to the degree. So the af number was never miscalculated, it was the LED-heat-stays-in-box case at a 25 C room with NEITHER assumption written down. The at figure WAS a genuine error, exactly as this run found. This run's 89-115 C Hoffman/Rittal figure is now recorded as the CONSERVATIVE BOUND and the ICD explicitly says do not average it. Budget: 10.5-12.5 W of box air at a 25 C room holding air to 70 C.
P3	PWM allocation NORMATIVE (ICD s3.5 rev A3): LUM-PAR-A runs PWM0-3 on ONE timer, 13-bit / 9.766 kHz, with a 90 degree phase stagger. CR-3 (14-bit/4.883 kHz) is GRANTED and available opt-in; CR-4 (stagger) is GRANTED.	The prior ICD claim that PWM0-3 sit on timer 0 and PWM4-7 on timer 1 is WITHDRAWN - the carrier does not hard-assign timers; each daughter declares its own map and firmware applies it from the ID_ADC code. Three timers remain free for this board. The 14-bit option doubles the driver timing budget from 141 ns to 281 ns and makes firmware dithering physically reproducible - it costs camera-flicker margin and must be re-tested against a phone camera on the first prototype.
P3	ADOPT the 14-bit / 4.883 kHz PWM mode (CR-3, granted) as this board declared timer map, instead of 13-bit / 9.766 kHz	It doubles the driver timing budget from 141 ns to 281 ns, which converts the shunt-FET timing risk from unverified-and-probably-violated into comfortable margin against the only switching number that exists (toff 150 ns typ). It ALSO doubles the LSB duration, which is what makes firmware PWM-domain dithering physically reproducible - spec-dimming s2.7 ranks 1 and 2 as a package, not alternatives. Cost is camera-flicker margin: band contrast rises from 0.61% to 1.23% at 60 fps (still under the visibility threshold) and 2.46% to 4.92% at 240 fps - and H1-P9 already recommends declaring 240 fps out of scope. ICD s3.5 makes the mode opt-in per daughter and requires a phone-camera re-test on the first prototype.

Phase	What	Why
P3	APPROVED library edits: J3 annulus 1.80->1.70; J4 annulus 1.60->1.70 and drill 1.050->1.100; inductor footprint swapped to KiCad stock Inductor_SMD:L_Cenker_CKCS4030; SMF15A accepted as-pulled with a note	J3/J4: ICD-01 s5.2 mandates 1.70 mm annulus on a 1.10 mm drill and daughters must conform - the carrier already normalised its mating DS1021 pair the same way (its lib/EDITS.md edit 1). Note the as-pulled J3 gap of 0.74 mm still CLEARS the 0.635 mm creepage requirement at 1.165x, so this is an ICD-conformance fix, not a safety fix - but conformance is the point of a frozen interface. Inductor: as-pulled land is 3.20 mm centres against a 2.60 mm datasheet land, a 0.60 mm error with pads hanging 0.40 mm past the body; KiCad stock at 2.70 is a strict superset of the datasheet land. SMF15A: 3.26 vs 3.00 pitch is a 0.13 mm out-board shift with pad size EXACTLY matching the datasheet - solderable, accept.
P4	P4 schematic-block agents run at opus/high instead of the tier table's fable/medium	Fable 5 credits exhausted account-wide. The skill's own tier table ranks fable/medium ABOVE opus/high, so opus is the nearest substitution that is not the silent downgrade the spawn rules forbid; user chose it over a cheaper mixed or all-sonnet split with the drivers/thermal safety content in view
P4	The power sheet owns ALL FOUR PWR.FLAGs (GND, +12V, +3V3, +48V_SW); every other sheet places consuming power symbols only	each rail enters the board through J3's passive pins on the power sheet, so that is the one honest driver; PWR.FLAG's pin is power_out and a second flag on the same rail collides power_out<->power_out and fails the erc gate
P4	J5 entry-direction question CLOSED: S10B-PH-SM4-TB is SIDE ENTRY and is the CORRECT part - no swap-part	datasheet p3 splits the model table into Top entry (B2B..B16B-PH-SM4-TB) and Side entry (S2B..S15B-PH-SM4-TB) columns, repeated twice more for the THT and LIF variants: the entry code is the LEADING letter, and the B in '10B' appears in every model of the series. Side entry is also right for this board - the module mounts to the enclosure wall so the harness leaves laterally regardless, and J5 is on B.Cu inside the ICD's 11.0 mm mezzanine gap where a top-entry PHR-10 housing plus bend radius would foul the carrier

Phase	What	Why
P4	The /PWM0..3 nets MUST carry the canonical root-level names /PWM0../PWM3; preferred mechanism is exposing them as hierarchical pins from control, with re-spelling constraints.json to /control/PWMn as the sanctioned fallback if that is not ERC-clean	sheets.md s2 states these are the names the schematic must produce, and constraints.json.high_speed feeds SEVEN consumers (check_return_path, planes_gen, route_critical, rules_gen, stitch_vias, constraints_lint, netlist_audit) - a mismatch silently drops the jitter-sensitive PWM nets from every one of them, on the board whose hardest requirement is the PAR-REQ-01 141 ns timing wall. It also fails netlist_audit outright (missing_net is an error), so it must be settled before the root build. sheets.md is internally inconsistent here - it lists PWM in the INTERFACE net table while its own From->to column calls them control-internal - so the tie-break is which artifact ends up TRUE: name-match is required either way, and the hier-pin route keeps one uniform convention across all root nets
P4	REVERSAL of the earlier P4 ruling: /PWM0..3 keep their real netlist names /control/PWM0..3, and constraints.json high_speed was re-spelled to match. The preferred hier-pin mechanism was tried, measured and REJECTED	the hier-pin route works electrically (netlist carried /PWM0 = J4-1 + R202-1 + U202-1) but is NOT ERC-clean: 4x label_dangling, because the root subgraph for those nets is wire + label + exactly ONE sheet pin and KiCad 10.0.3 requires a real connection point. Two probes settled the rule: a second same-named root label does NOT satisfy it (8 errors, both flagged), while one TestPoint pin per root PWM wire takes the identical project to 0/0 - so buying the bare name would mean adding four PARTS to the root, a design change rather than a stitch fix. Every other root net escapes this only because two children expose it. lumina-carrier never exercised it: all 70 of its root labels appear 2-3 times. Outcome is equivalent for the seven high_speed consumers, which match on exact name and do not validate name shape

Sheet plan

lumina-par (LUM-PAR-A) - hierarchical sheet plan

Five sheets under a root that contains nothing but the sheet symbols and the inter-sheet wiring. Interface nets become hierarchical pins at P4 and placement groups at P6.

Net-naming contract (binding on P4). KiCad names a net from a hierarchical label alone as `<sheet>/<LABEL>`. To produce the canonical names below, ****P4** must place a root-sheet local label on every inter-sheet wire**, spelled exactly as in s2. `constraints.json` uses these names and P5-P8 will silently no-op on any net whose name does not match. Power nets are bare global symbols (`+3V3`, `+12V`, `+48V_SW`, `GND`); every other inter-sheet net carries a leading `/`.

1. Sheets, refdes ranges and contents

Refdes are unique across sheets by construction: each sheet owns a hundreds block for every prefix, and each sheet has its own `#PWR` base.

| Sheet | Blocks | Refdes block | ‘#PWR’ base |

```

|---|---|---|---|
| 'power' | B1 - rail entry, bulk, branch-B front end (DNP), test points, H5 |
**100-199** | **100** |
| 'control' | B2 - ENABLE gating and fail-safe logic; B5 - ID and calibration |
**200-299** | **200** |
| 'drivers' | B3 - the four constant-current channels and their shunt FETs |
**300-399** | **300** |
| 'thermal' | B4 - NTC dividers, window comparator, FAULT driver | **400-499** |
**400** |
| 'led_if' | B6 - LED harness header, TVS clamps, thermocouple pads | **500-599**
**500** |

```

Exception: J3, J4 and H5 keep the refdes the ICD assigns them (ICD s7.2 / s7.5 name them J3, J4 and H5, and the carrier's placement tables refer to them by those names). They are exempt from the hundreds blocks. J5 is this board's own harness header and follows the same connector sequence.

1.1 power (100-199, #PWR 100)

```

| Ref | Part | Note |
|---|---|---|
| **J3** | CONNFLY DS1023-2*7SF11, 2x7 socket | **reverse-mounted, bottom side,
facing down** (ICD s7.3). Pin 1 silkscreened with a triangle |
| **H5** | MountingHole_3.2mm_M3 | (46, 74), ICD s7.5. Not a 'board_init' hole -
added here so it carries a refdes |
| C101, C102 | 22 uF / 25 V X7R | '+12V' bulk |
| C103, C104 | 10 uF / 16 V X7R | '+3V3' bulk |
| C105 | 100 nF | '+3V3' HF |
| TP101-TP103 | test pads | '+12V', '+3V3', '+48V_SW'. **TP103 carries the ICD
s9 bench-hazard silkscreen** |
| **Q101, Q102, R101-R104, C106-C108** | **branch-B front end - DNP on branch
A** | 100 V P-FET hot-swap, Cgd dV/dt network, gate pull-down N-FET from
'/EN_OK', 100 k **0805** bleed, 2x 10 uF/100 V bulk. R104 is 0805 now so
branch B needs no footprint change (ICD s5.4) |

```

1.2 control (200-299, #PWR 200)

```

| Ref | Part | Note |
|---|---|---|
| **J4** | CONNFLY DS1023-2*12SF11, 2x12 socket | reverse-mounted, bottom side,
facing down |
| U201 | SN74LVC1G08 (SOT-23-5) | '/EN_OK = ENABLE AND FAULT' |
| U202 | 74LVC00A quad NAND (SO-14) | '/SHUNTn = NOT(PWMn AND /EN_OK)' - one
package covers all four channels |
| U203 | 24C32-class I2C EEPROM | **address 0x50**, WP to GND. **No I2C pull-ups
on this board** |
| R201 | 100 k | **ENABLE pull-down at the connector end, before any series
element** (ICD s8.2) |
| R202-R205 | 4x 100 k | PWM0-3 pull-downs at J4 - an undriven carrier must not
float a gate input high |
| R206 | **VALUE TBD** | 'ID_ADC' bottom leg. **Allocated by the carrier owner,
not chosen here** (ICD s3.3). P4 blocker - 'decisions.md' OPEN-2 |
| R207 | 100 k | local 'FAULT' pull-up to '+3V3', so the node is defined when
unmated |
| R208, R209 | 0 R / 10 k | EEPROM A0-A2 and WP strapping |
| R214-R217 | 4x 0 R | '/EN_OK -> /DRV_ENn' links, **fitted by default** |
| **U204, R210-R213, C210-C213, D201-D204** | **converter-idle one-shot - DNP
option** | 74LVC14 hex Schmitt + retriggerable RC + diode per channel. Populate
to save ~0.8 W at saturated colours ('power_tree.md' s2); if populated, remove
R214-R217 |

```

| C201-C205 | decoupling + 'ID_ADC' 100 nF | |

1.3 drivers (300-399, #PWR 300) Four identical channels; channel n is based at 300 + 20n. Channel 0 shown; channels 1-3 are U321/U341/U361 and so on.

Ref	Part	Note
U301	TPS92515HVDGQR class (MSOP-10-EP)	5.5-65 V, 2 A internal FET, shunt-PWM capable. Exposed pad on a >= 9-via array
L301	47 uH shielded	sized for **ripple, not slew** - 90 mA (30 %) at ~700 kHz
D301	**60 V** Schottky, 1 A	branch A. Low Vf matters: it dominates the 0.20 W idle loss. Branch B needs a 100 V part
R301	sense resistor	sets 300 mA (af, branch A). **The at upgrade is this value**
Q301	60 V logic-level N-FET	shunt across the string. **60 V now, not 30 V**, so the branch-B / 'at' 8S string (27.2 V) needs no change
R302	150 R	shunt gate series - bounds the 74LVC00 output to 22 mA, ~30 ns edges
R303	100 k	shunt gate pull-down - an unpowered gate IC must not leave a FET half-on
C301	4.7 uF / 25 V	VIN local (100 V on branch B)
C302	100 nF	VIN HF, at the pin
C303, R304	**DNP**	SW-node RC snubber footprint

There is no output capacitor across any LED string - a shunt FET dumps it every PWM cycle (blocks.md B3 rule 1). Do not let a reviewer add one.

1.4 thermal (400-499, #PWR 400)

Ref	Part	Note
U401	quad open-drain comparator, LMV339 / TLV3704 class	rail-to-rail input, <= 50 uA/channel. **Open-drain outputs are mandatory** - they wire-OR onto 'FAULT', which must never be driven high
RT401	10 k B3950 NTC, 0603	**on the copper of the hottest driver stage, outside the DC-DC hot zone (2,46)-(36,68)**
R401, R403	2x 10 k	NTC divider top legs (board, module)
R402, R404	<= 1 k	ADC series - keeps total source impedance <= 6 kohm against the ICD's 10 kohm
R405-R412	reference ladder + hysteresis	ratiometric off '+3V3', so thresholds track the carrier's ADC reference
C401-C403	decoupling + NTC filter	filter caps sized so R402/R404 x C stays a slow-signal filter, not a pulse filter

Four comparator channels: **emitter hot (90 C, release 75 C) / emitter open / emitter short / board hot (110 C, release 95 C)**. The window (open + short) converts the most likely fault in an off-board module - a broken harness wire, which reads as *cold* in either divider orientation - from fail-dangerous to fail-safe (D-T17, E-10).

1.5 led.if (500-599, #PWR 500)

Ref	Part	Note
J5	10-way latched wire-to-board, 2.0 mm (JST PH class)	4 anode + 4 per-channel GND return + '/NTC_LED' + dedicated NTC sense return
D501-D504	TVS, SOD-123	**15 V standoff on branch A, 33 V on branch B** - same footprint, a BOM value change
TP501, TP502	bare-copper thermocouple pads	bring-up thermal verification (L-13). ICD s9 bench-hazard silkscreen

No 48 V net reaches this sheet, ever (D-T13). That is what keeps the 0.635 mm clearance rule, the 100 V capacitor rule and the 0805-minimum resistor rule off the harness and the module.

2. Interface nets - canonical names

These are the names the schematic **must** produce. `constraints.json` and every P5-P8 check key off them.

Net	Type	From ->	to	Note
'+48V_SW'	power	'power'	(J3 1/3/5)	**landed, not tapped** on branch A. Declared in 'voltages' at 57 V so 'check_creepage' enforces the clearance
'+12V'	power	'power' ->	'drivers'	0.717 A max ('power_tree.md' s3, revision B)
'+3V3'	power	'power' ->	'control', 'thermal'	<= 5 mA
'GND'	power	everywhere	In1 plane	
'/PWM0' .. '/PWM3'	in	'control'	(J4) internal	3.3 V CMOS. **No RC filter; if any network, tau <= 14 ns**
'/ENABLE'	in	'control'	(J4) internal	100 k pull-down here
'/FAULT'	bidir, open drain	'thermal' ->	'control' ->	J4 **never driven high** ; wire-OR'd with the carrier eFuse fault
'/EN_OK'	internal	'control' ->	'power'	(branch-B gate) 'ENABLE AND FAULT'
'/DRV_EN0' .. '/DRV_EN3'	out	'control' ->	'drivers'	driver PWM/EN pins. Tied to '/EN_OK' through 0 R by default; driven by the one-shot if populated
'/SHUNT0' .. '/SHUNT3'	out	'control' ->	'drivers'	shunt-FET gates. 'NOT(PWMn AND /EN_OK)'
'/LED0.A' .. '/LED3.A'	out	'drivers' ->	'led_if'	channel anodes, 0.30 A each
'/NTC_LED'	in	'led_if' ->	'thermal'	module NTC divider node
'/NTC_BRD'	internal	'thermal'	board NTC divider node	
'/ADC0'	out	'thermal' ->	'control' ->	J4 emitter temperature to the carrier. Source impedance <= 6 kohm
'/ADC1'	out	'thermal' ->	'control' ->	J4 board temperature to the carrier
'/ID_ADC'	out	'control' ->	J4	divider bottom leg; value allocated by the carrier owner
'/I2C_SCL', '/I2C_SDA'	bidir	'control'	(J4) ->	U203 **no daughter-side pull-ups**

Unused ICD signals, landed on J4 and left unconnected on this board: PWM4..PWM7 (no fifth channel on rev A), DSPI_SCK/MOSI/MISO/CSn (a 4-channel PWM design needs no serial device beyond the EEPROM, and `decisions.md` D2 rejects a local PWM generator). Leave them as no-connects with an explicit ~ ERC directive so P4's ERC does not flag them and so a reviewer can see the decision.

I2C address map (this board owns the whole space; the carrier reserves nothing): ****0x50 = 24C32 EEPROM; 0x51-0x57 reserved for its own address pins; nothing else on the bus.**** The comparator and both NTCs are analogue.

3. Placement groups for P6

Declared in `constraints.json` under `placement.groups`. Each buck channel is one tight cluster; the loop that matters is +12V -> C301 -> U301 -> L301 -> D301.

Group	Anchor	Members
---	---	---

'ch0'	U301	L301, D301, R301, R302, R303, C301, C302, Q301
'ch1'	U321	L321, D321, R321, R322, R323, C321, C322, Q321
'ch2'	U341	L341, D341, R341, R342, R343, C341, C342, Q341
'ch3'	U361	L361, D361, R361, R362, R363, C361, C362, Q361

'ntc_brd' | U301 | RT401 - the board NTC must sit ****on**** the hottest driver stage (T-3) |

'gate' | U202 | R202-R205, C202 - keep the NAND next to J4's PWM pins to keep the jitter-sensitive run short |

'analog' | U401 | R401-R412, C401-C403 - ****separated** from every inductor by **>= 10 mm**** |

placement.fixed: **H5**, positioned by an explicit place_edit op at (46, 74) in board-local coordinates (translated - see stackup.md TRAP 2).

placement.edges:

Ref	Edge	pos	Why
J3	bottom	0.24	ICD s7.2 nominal centre x = 24 mm. **Hint only**
J4	bottom	0.72	ICD s7.2 nominal centre x = 72 mm. **Hint only**
J5	**top**	0.56	x ~ 56 mm - the only clear span of the top edge (the notch occupies x 6-36, the recovery-header keepout x 76-98), and it sits directly above the driver band so the LED anode runs stay short

J3 and J4's entries are hints to the annealer, not final positions. Both must be placed with explicit place_edit ops at the re-issued ICD s7.2 coordinates and then **locked**, before the annealer runs. Both are **bottom-side, facing down** - set the side at P4 in the footprint or with a place_edit flip op, and check pin 1 in the mated view, not from the footprint (pin numbering mirrors across the row on a bottom-side part).

4. Sheet-level P4 notes

1. **The DNP option sets must be in the netlist**, not added later: the branch-B front end (power Q101/Q102/R101-R104/C106-C108) and the converter-idle one-shot (control U204/R210-R213/C210-C213/D201-D204). A DNP part still has pads, so its clearance and area are accounted for at P6/P7 and the option stays a populate change rather than a respin. Mark them DNP in the BOM variant field so bom_cpl at P9 excludes them.
2. **'FAULT' is open drain.** ERC will want a driving pin somewhere; the comparator outputs plus R207 are it. Do not let a fix loop add a push-pull driver.
3. **No I2C pull-ups.** Fitting them is an ICD violation, and P4's reflex is to add them. Record it on the schematic.
4. **'PWM4..7' and 'DSPL'* are deliberate no-connects.
5. **R206's value is unknown until the carrier owner allocates it.** Place the part with a TBD value field and a schematic note; do not guess a value - that is exactly the silent divergence ICD-01's preamble forbids.

5. P4 AMENDMENTS (2026-08-07)

This file was written at P2. The items below were settled during P4 schematic capture and its adversarial review, and **they supersede the sections above.** architecture/p4-wiring-notes.md remains binding for wiring rules.

5.1 s1.2 control - new parts

Ref	Part	Note
R218-R221	4x **10 k**	'/DRV_EN0..3' fail-safe pull-downs, one per channel, on the '/DRV_ENn' side of the R214-R217 links. **10 k is load-bearing

- do NOT "harmonise" these to the board's usual 100 k.** The TPS92515HV enable pin sources up to 25 uA, so 100 k sits at 2.5 V, above the 1.0 V threshold. Without them, every power-up window with '+12V' up and '+3V3' not yet (and any 12 V-only bench bring-up) leaves all four drivers undriven and latching ON - a full-current LED flash, defeating ICD s8.2. Placed with their DRIVER, not with the links. |

R206 is **4.7 k 1 %** (CR-1), not "VALUE TBD". parts.json is re-issued.

5.2 s1.3 drivers - new parts and one changed value The COFF network (C304/R305/R306 per channel, s1.3 had none of it) is joined by the **BOOT network, which s1.3 also omitted and without which the converter cannot switch at all**:

Ref (ch0; +20/channel)	Part	Note
--- --- ---		
C305	100 nF	BOOT capacitor at pin 4 - the high-side FET's gate supply
D302	1N4148W	BOOT diode, anode on the device's own VCC (pin 2), cathode on BOOT
C306	4.7 uF	VCC decoupler at pin 2, datasheet sec 8.3.6

ROFF2 (R306 etc) is 30 k, NOT 47 k. The original 47 k was sized from TI Equation 9, which assumes VCC charges COFF through ROFF2 alone - false at ROFF1 = 10 k. The shunted COFF node then asymptotes **below** the 1.00 V VOFT at every corner, the off-timer never trips, and shunt-dimming linearity is lost - the exact Figure-43 failure the network was added to prevent. 30 k is solved jointly (dIL_shunt == dIL_normal); 0 of 243 corners now fail. **Do not re-derive this from TI's text alone**: Eq 1 and Eq 8 disagree about the freewheel diode drop and lead to an unsafe ~38 k.

5.3 s1.4 thermal - range extended, values changed

- Ladder + hysteresis is **R405-R416**, not R405-R412.
- **R402/R404 are 150 R, not " ≤ 1 k"**, and C402/C403 are **4.7 uF 0805, not 100 nF 0603** (footprint change) so the filter stays slow at the lower R.
- **R413/R415** (10 k) series-protect the comparator inputs and **D401** (1N4148W) clamps /NTC_LED to +3V3. The LM339LV inputs are diode-clamped to GND **only**, with no clamp to V+, so a series resistor alone does not protect them from a harness fault that puts an LED anode (6.8 V nominal, 16.7-24.4 V at the TVS clamp) onto the sense line.
- Hysteresis is returned to the **protected comparator nodes**, not the sensor dividers. See the blocks.md B4 amendment for the solved thresholds.

5.4 s2 net table - /PWM0..3 correction The s2 table's '/PWM0'..'/'PWM3' spelling is not what the netlist produces. Those nets are control-internal and come out '/control/PWM0..3'; constraints.json.high_speed was re-spelled to match, so all seven high_speed consumers bind correctly. Promoting them to root-crossing names was tried and measured: it is electrically correct but raises 4x label_dangling, because such a net reaches the root as wire + label + exactly ONE sheet pin, and buying the bare name would mean adding four PARTS to the root. /ENABLE, /ID_ADC, /I2C_SCL and /I2C_SDA are in the identical position and are also /control/NAME. No constraint references those four.

Full architecture narrative (not inlined; see the artifact index): architecture/blocks.md, architecture/decisions.md, architecture/power_tree.md, architecture/stackup.md.

5 Schematic

Gate **erc** (phase P4): **pass** after 2 attempt(s); last 2026-08-07T13:18:16, failing 0 of 0. The full schematic PDF follows.

file: power.ttest.pdf

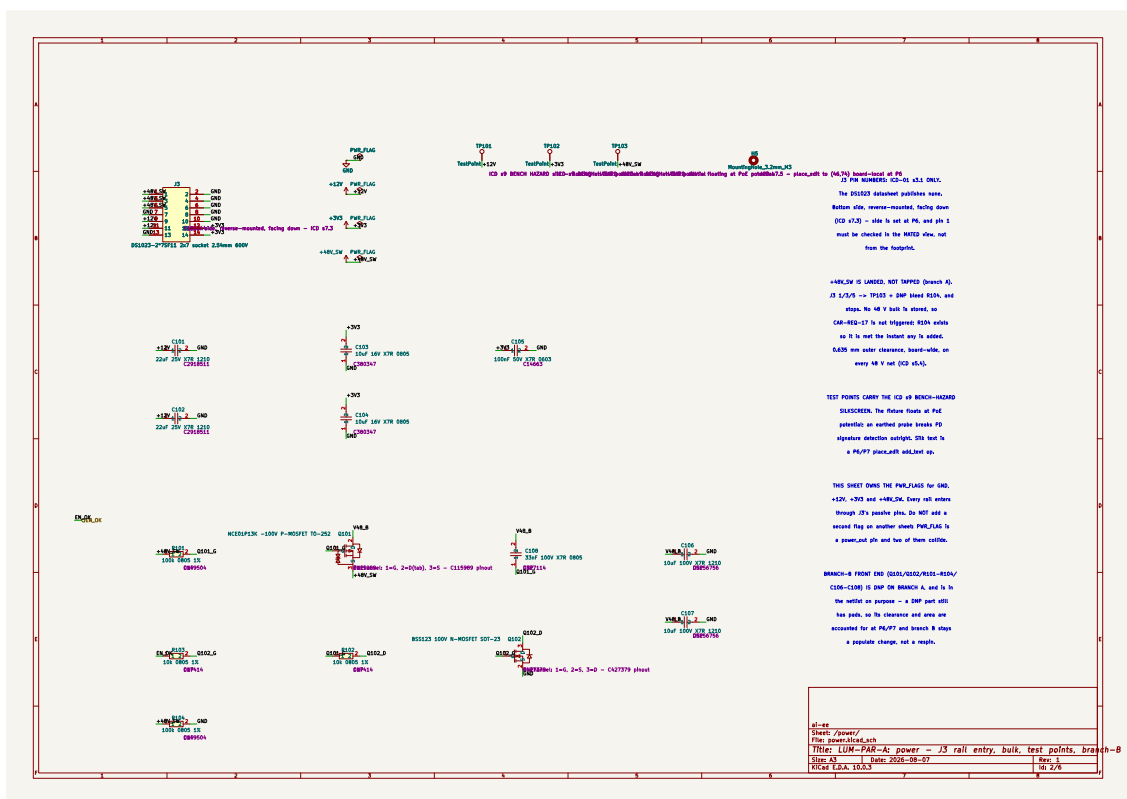
File: thermalfood_ach

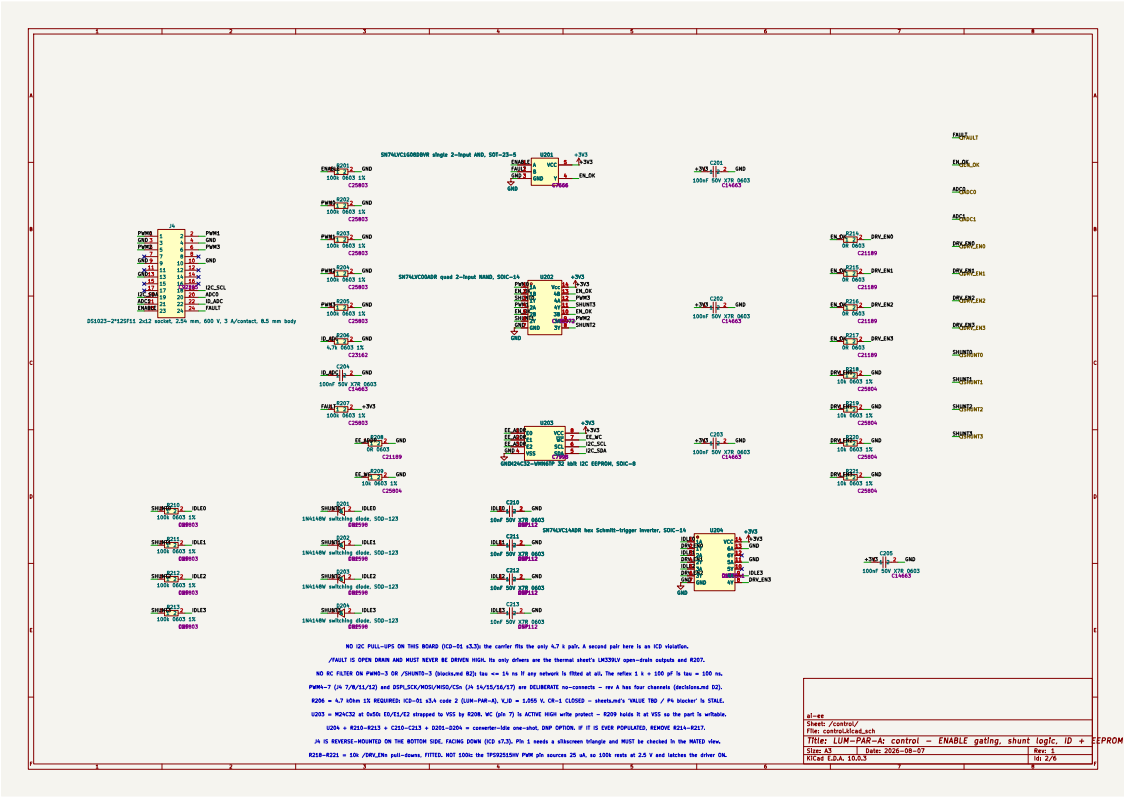
File: control_hvac_act

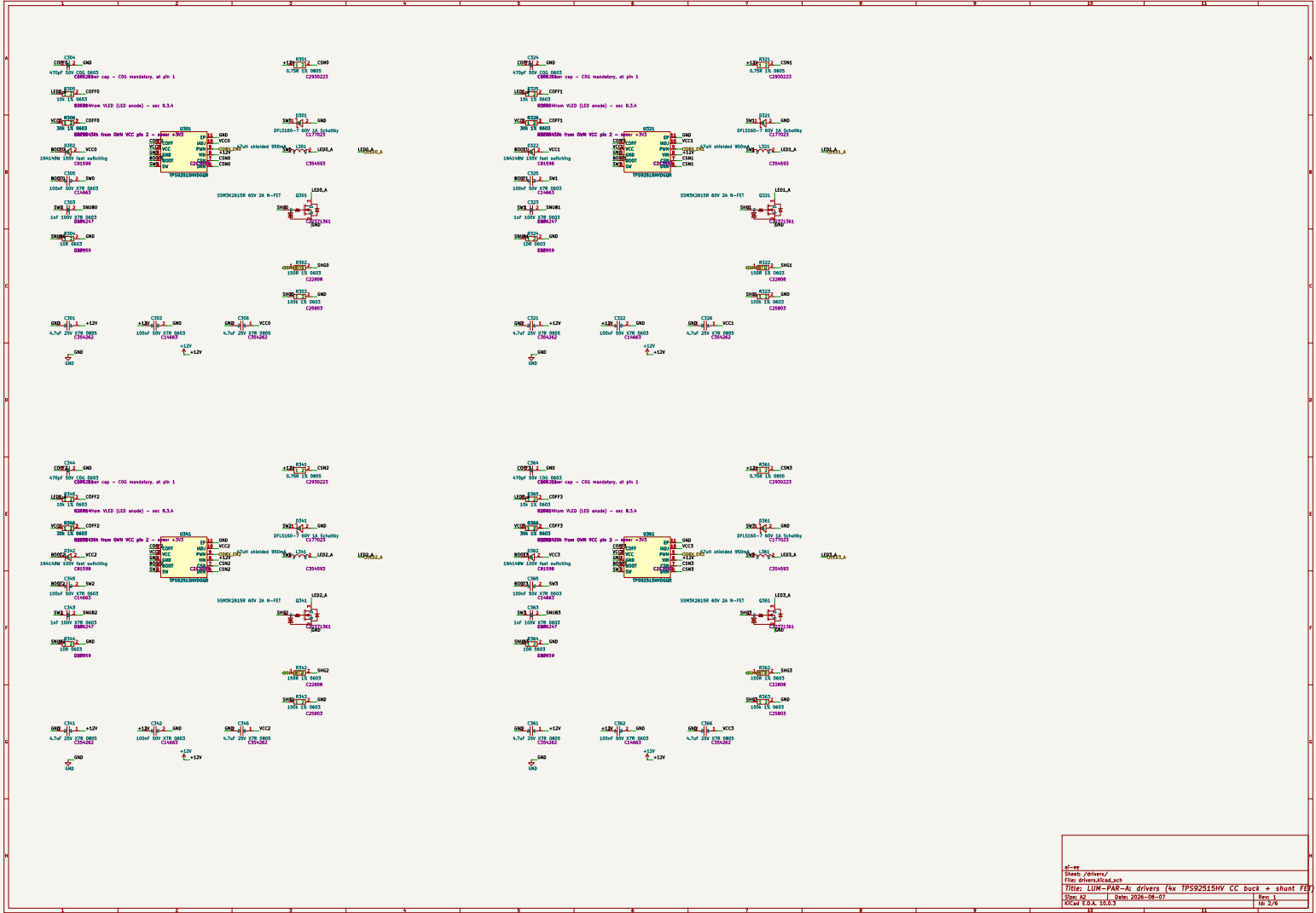
Flieg: led, if, ahead, sch

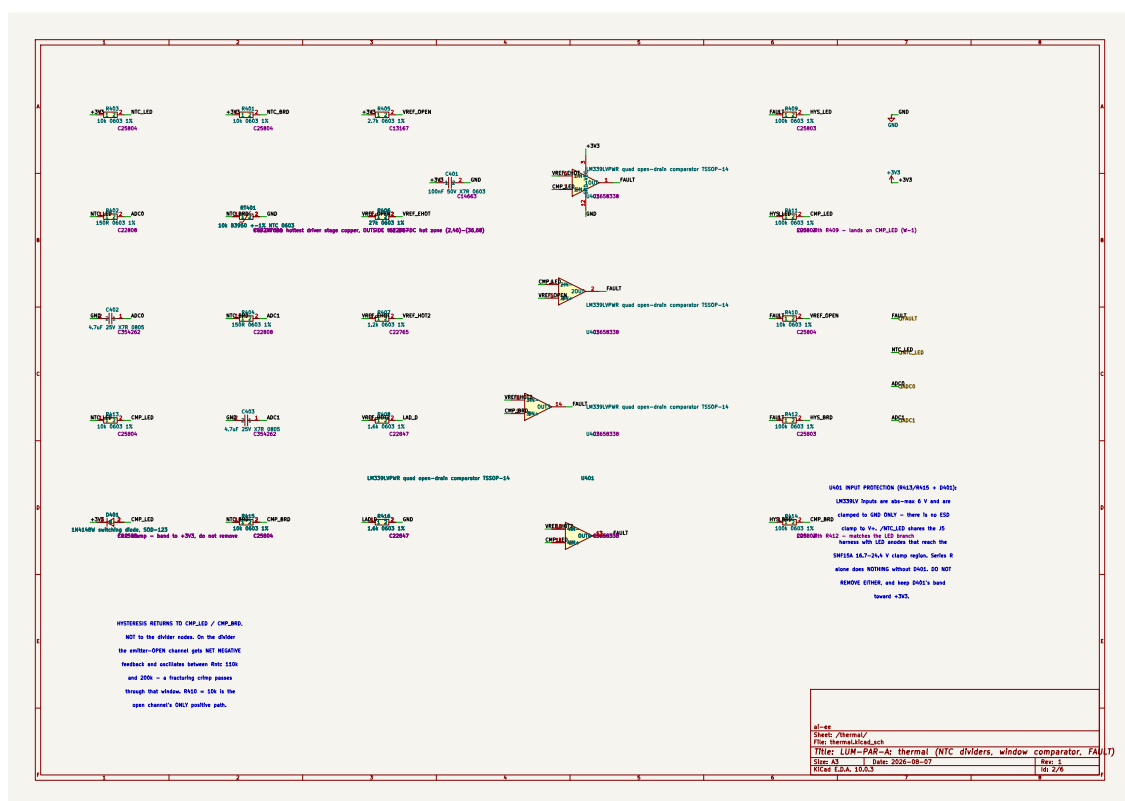
File: delvans-ikind sch

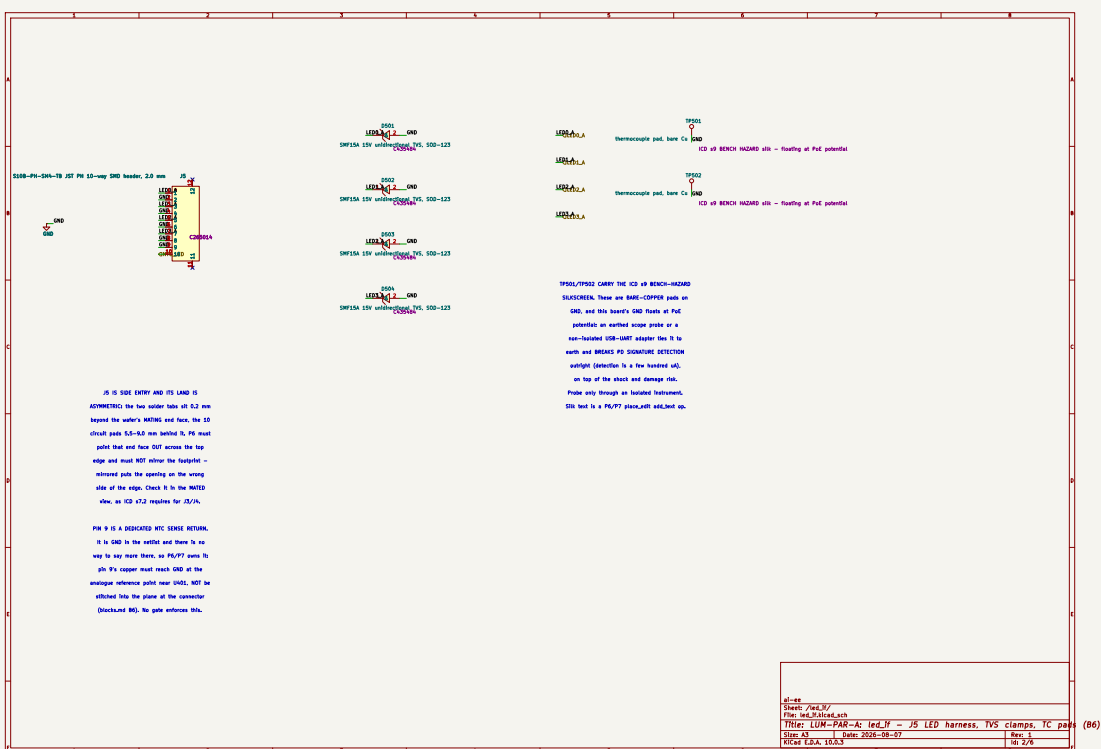
Rev: 1











6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake digest - lumina-par (LUM-DTR-PAR-A / LUM-PAR-A)

- Board: RGBW par daughter for LUM-CAR-A. 4 (maybe 5) PWM-dimmed
constant-current LED
channels, stacked 11.0 mm above the carrier on the frozen ICD-01 connector
pair.
Continuous duty; the design problem is dimming quality and fixture-to-fixture
colour
consistency, not peak power.
- Inherited as CLOSED (05-lumina-closed-decisions.md): D-01 (at-capable stage,
af-classified;
budget 8.6-9.3 W af / 18.7-20.0 W at to this daughter - the "~8.5 W" and "~10 W
regulated"
figures in briefs 00/03 are dead), D-02 (48 V + 12 V + 3.3 V on the connector),
D-03 (no UV).
- ICD-01 is a frozen hard input. All interfaces are ICD-only: J3 2x7 power,
J4 2x12 signal.
ICD s9 forbids adding an external connector of any kind. Common footprint 100
x 80 mm,
R3.0, 5x M3, 1.6 mm, plus a mandatory 30 x 26 mm RJ45 notch at (6,0)-(36,26).
- Safety flags: >30 V unconditional (48 V present on J3 whether tapped or not
- 0.60 mm outer
clearance, 100 V caps, 0805+ resistors across the 48 V domain); whole fixture
floats at PoE
potential including LED wiring and any heatsink; >3 A conditional on
per-channel LED current
that P1/P2 must publish.
- Riskiest unknowns, in order: (1) af-vs-at sizing - +12V tops out at 15.0 W
so at-sizing forces
a >=60 V stage on THIS board, the exact thing D-02's 12 V rail existed to
avoid; (2) continuous
thermal in a non-conductive enclosure at 56-69 C internal air; (3) PAR-REQ-01's
"5-10 % of full
output" is ambiguous and changes the required driver settling time by ~40x.
- Hard derived requirement recorded: the board must be electrically incapable
of exceeding the
carrier's fault ceilings (12 V OCP 2.0 A, 48 V eFuse 1.0 A latch-off) with
every PWM stuck at
100 %, because the PAR-REQ-11 clamp is firmware and firmware hangs.
```


- Pipeline risk logged for P5: 'board_init.py' has no notch/cutout option and 'kc.py' has no outline editor, so the mandatory RJ45 relief needs a direct Edge.Cuts edit after board_init; and '--mounting-holes' places holes at margin/2, so the ICD's 5 mm inset needs '--margin 10'.
- Artifact: 'requirements.md' (9 sections, 16 open questions, all with recommendations).

P1

P1 Research digest - lumina-par

Roster: 2x component-scout (LED driver, RGBW emitter), 1x reference-design (continuous thermal + protection chain), 1x interface-spec (PWM dimming / flicker / perceptual resolution). No power-architect: the rail tree is fixed by ICD-01 s6.2 and the only open variable was which rail feeds the LED stage, which became conflict C1 for the architect.

- ****Thermal is the binding constraint, not the power budget.**** A sealed 120x100x60 mm non-metallic box is 3.6-4.3 K/W internal-air-to-room (Hoffman ~4.6 W/m2K, cross-checked vs Rittal k=5.5). Nothing on the PCB changes that number. Sealed caps total box heat at ~5.0 W for 45 C internal air in a 25 C room; minus the carrier's 2.4 W that leaves ~2.6 W of daughter heat = ~3-4 W of LED electrical power, ****under half the af envelope****.

- ****ICD s7.6's internal-air figures do not hold.**** Its 69 C (at) is internally inconsistent with its own 56 C (af): convection is near-linear in dT at this scale, so ~2x the heat must give ~2x the rise (31 K -> ~60 K -> ~85 C), and the

independent calculation gives 89-115 C. Optimistic by 20-46 K. Raised as a blocking issue against LUM-CAR-A under ICD s10 (OPEN-1).

- ****PAR-REQ-01 at 5 % perceived fails at 13-bit by 8.9x**** and needs ~17 bits; ESP32-S3 LEDC caps at 14. Two independent walls: resolution (17 bits) and driver settling ($T \leq 141$ ns). New fact, verified from ESP-IDF source: LEDC does

NOT rigidly couple frequency to resolution - "80 MHz / 2^n " is a maximum, so 14-bit is available well below 4.88 kHz. Widens the trade space.

- ****Flicker is over-specified.**** IEEE 1789-2015 RP2 removes all modulation-depth

restriction above 3 kHz; 9.766 kHz sits 3.3x above it. The 10 kHz-over-1.2 kHz camera call is confirmed (8.1x lower band contrast) but bounded: clean to 120 fps, marginal at 240 fps, 960 fps unreachable at any usable LEDC setting.

- ****No CC LED driver has an enable pin separate from its PWM pin.**** On all five surviving candidates the dimming pin IS the enable, so ENABLE must be gated with

each PWM line externally. The brief's "driver with an EN pin strongly preferred"

is not achievable in this performance class.

- ****Emitter is single-source.**** The XINGLIGHT 4-in-1 is the only in-stock RGBW power emitter above 50 mA on the whole LCSC catalogue (318-part sweep; every Cree XLamp colour line at 0 stock), with no published Rth and no published Tj max. Fallback is RGB 3-in-1 + separate white at the cost of PAR-REQ-15.

- Orchestrator correction applied to 'requirements.md': s9 Q9(b) on-times were 10x

high (1.4-6.1 us -> 141-646 ns), verified by independent recompute.

Load-bearing:

it moved the strict reading of PAR-REQ-01 from "several drivers manage it" to "no surveyed CC driver manages it by gating the converter".

P2

P2 Architecture digest - lumina-par

Artifact: 'architecture/' (blocks.md, power.tree.md, stackup.md, sheets.md, constraints.json, decisions.md). This is the H1 checkpoint package.

- ****6 blocks****: rail entry (48 V landed on J3 but NOT tapped), ENABLE gating, 4x TPS92515HV-class CC buck with a shunt FET per channel, window-comparator over-temperature protection, ID divider + I2C EEPROM, 10-way LED harness. Emitters off-board on an aluminium MCPCB.
- ****Stackup JLC04161H-3313, 4 layers****, 100.0 x 80.0 mm, 1.6 mm. Drivers: switch-node containment 11 mm above a live 2.4 GHz antenna, a continuous GND reference under jitter-sensitive PWM, B.Cu consumed by two reverse-mounted sockets, ~120 parts in ~45 cm². No controlled-impedance net exists on this board.
- ****Design point 150 mA/die, 4 packages, 2S2P per channel.**** Full white (the stuck-PWM hardware backstop) = 0.718 A on +12V: 4 % under the sustained ceiling and 2.8x under OCP. The board is electrically incapable of exceeding its budget.
The >3 A conditional flag is CLOSED - max per-channel current is 0.30 A.
- ****C1 arbitrated to +12V****, and the P1 argument for 48 V was withdrawn rather than overruled. The "48 V gives 16x faster slew" claim held the inductor fixed across a 4x change in (Vin - Vs). With L sized for the same ripple fraction, $t_r + t_f = 1/(f \cdot k)$ - the rail voltage cancels out of the algebra. Verified independently by the orchestrator: 4.762 us on BOTH rails at 700 kHz / $k=0.30$ (48 V only redistributes 3.81/0.95 us between rise and fall). That is 33.8x the 141 ns budget, so no rail or inductor choice reaches PAR-REQ-01 by gating the converter.
- ****C2 resolved: shunt-FET dimming closes the timing wall (T ~ 50-100 ns vs 141 ns, 1.4-2.8x margin); the resolution wall is NOT closable on this board**** and needs carrier firmware dithering (CR-5). spec-dimming's "local >=16-bit PWM IC" class LOSES to led-driver's six-part survey of that exact class - no part delivers >=150 mA/ch AND >=10 kHz AND useful resolution AND independent channels.
- ****Cost ~\$18-23/board**** delivered (BOM ~\$12.9, driver silicon 52 %) against a \$25-35 target; module + heatsink adds \$8-14/fixture.
- ****Sealed enclosure does not close**** with the module inside (9.15 W -> 62 C air against a 45 C criterion). It closes vented, or sealed with LED heat conducted through the wall (3.21 W -> 38 C). Seven ENC-x acceptance criteria written for whoever owns the enclosure, since it is not this board's deliverable.
- Riskiest decision: ****emitters off-board on MCPCB (D3)**** - the thermal argument rests on an ASSUMED 12 K/W package Rth, because the vendor publishes neither Rth nor Tj max, and it is the only in-stock RGBW power emitter on LCSC. At 20 K/W even MCPCB is marginal at af.
- Two P5 traps recorded: fixed-outline board.init derives its origin from the packed component bbox (every rect in constraints.json needs a P5 translation), and the antenna column has no automated checker on F.Cu/B.Cu.

P3

```
# P3 Parts + Library digest - lumina-par

Artifact: 'parts/parts.json' (47 lines), 18 datasheet extracts, 'lib/' (91 sym
/
23 fp), 'reports/fp_verify_*'. Ran 2026-07-28; digest written at the resume.

- Emitters 4x C22434861 RGB + 4x C48586656 white, 2S2P, 150 mA/die.
- **OPEN-8 REFUTED**: no 140/120 deg beam mismatch - both datasheets say 140,
the
LCSC record is wrong. Diffuser spec and its 30-45 % flux cost must be
re-derived.
- Sense 0.75 ohm (275 mA/ch); shunt FET SSM3K2615R (C22371361).
- **COFF network was MISSING, now fitted**: 12 parts, ROFF1 = 10k not TI's
8.2k;
without it the shunt topology loses dimming linearity (TPS92515HV 8.3.4).
- CR-1 closed: R206 = 4.7 k 1 % (supersedes sheets.md "TBD"). CR-3 adopted:
14-bit / 4.883 kHz PWM. CR-6 closed by ICD s7.7 A5. Emitter Rth still absent.
- fp_verify: 4 violations, all adjudicated (2 off-board emitter modules,
thermal
pad; L301/D501 approved KiCad-stock land deltas).
- **To H2**: shunt-FET switching UNVERIFIED at 3.3 V (toff 150 ns TYP > the
141 ns
pulse; both gate-path ends ungrounded -> P8 bench); compounded light budget;
power_tree.md quotes 300 mA where the board builds 275 mA.
- Tooling defects (central): lib_pull false-pass, fp_verify blind to drill,
placelib ignores pad rotation -> hand to P6.
```

P4 (no digest recorded)

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-07T11:58:05	erc	pass	0/0
2026-08-07T13:18:16	erc	pass	0/0

Issues

Id	Phase	Kinds	Severity	Status
1		coff-shunt-timer-dead, drv-en-undriven-no-3v3, hysteresis-on-sensor-node, missing-bench-hazard-note, ntc-harness-input-unprotected	error	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-07-28T20:03:50	H1 approved with 4 directed changes: (Q1) enclosure sealed + LED heat conducted through the wall; (Q2) emitter changed to RGB 3-in-1 + separate white, AGAINST the run's recommendation, chosen knowingly for a published 125 C Tj and 3.5x stock over an assumed 12 K/W Rth from a sole source; (Q3) firmware commits to PWM-domain dithering ≥ 3 -4 bits, strict gamma-2.2 reading of PAR-REQ-01 binds; (Q4) same-reel sourcing, 24C32 fitted but shipped empty, PAR-REQ-16/17 to be formally amended.
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
state.json	74,449 B	
kicad/lumina-par.kicad.sch	21,586 B	
requirements.md	42,639 B	
architecture/blocks.md	36,027 B	
architecture/constraints.json	16,716 B	
architecture/decisions.md	63,672 B	
architecture/p4-wiring-notes.md	6,439 B	
architecture/power_tree.md	36,065 B	
architecture/sheets.md	15,823 B	
architecture/stackup.md	45,870 B	
reports/fp_(module)_C22434861.overlay.svg	2,952 B	
reports/fp_(module)_C48586656.overlay.svg	1,512 B	
reports/fp_courtyard_adequacy.json	3,018 B	
reports/fp_D201_C81598.overlay.svg	1,128 B	
reports/fp_D301_C177023.overlay.svg	1,146 B	
reports/fp_D501_C435484.overlay.svg	1,131 B	
reports/fp_icd_annulus.json	2,010 B	
reports/fp_J3_C113344.overlay.svg	5,550 B	
reports/fp_J4_C92265.overlay.svg	9,250 B	
reports/fp_J5_C265014.overlay.svg	4,809 B	
reports/fp_L301_C354593.overlay.svg	1,130 B	
reports/fp_Q101_C115989.overlay.svg	1,514 B	
reports/fp_Q102_C427379.overlay.svg	1,513 B	
reports/fp_Q301_C22371361.overlay.svg	1,519 B	
reports/fp_U201_C7666.overlay.svg	2,237 B	
reports/fp_U202_C485072.overlay.svg	5,590 B	
reports/fp_U203_C7998.overlay.svg	3,353 B	
reports/fp_U204_C133541.overlay.svg	5,590 B	
reports/fp_U301_C213553.overlay.svg	4,415 B	
reports/fp_U401_C3658338.overlay.svg	5,517 B	
reports/fp_verify_(module)_C22434861.json	2,488 B	
reports/fp_verify_(module)_C48586656.json	2,806 B	
reports/fp_verify_D201_C81598.json	1,892 B	
reports/fp_verify_D301_C177023.json	2,273 B	

File	Size	Note
reports/fp_verify_D501-C435484.json	2,827 B	
reports/fp_verify_J3-C113344.json	2,036 B	
reports/fp_verify_J4-C92265.json	2,035 B	
reports/fp_verify_J5-C265014.json	2,417 B	
reports/fp_verify_L301-C354593.json	3,481 B	
reports/fp_verify_Q101-C115989.json	2,311 B	
reports/fp_verify_Q102-C427379.json	3,137 B	
reports/fp_verify_Q301-C22371361.json	3,277 B	
reports/fp_verify_summary.json	9,032 B	
reports/fp_verify_U201-C7666.json	1,274 B	
reports/fp_verify_U202-C485072.json	1,827 B	
reports/fp_verify_U203-C7998.json	1,330 B	
reports/fp_verify_U204-C133541.json	1,844 B	
reports/fp_verify_U301-C213553.json	2,429 B	
reports/fp_verify_U401-C3658338.json	1,290 B	
reports/lib_pin_types.json	20,659 B	
reports/lib_pull.json	6,069 B	
reports/netlist_audit.json	3,457 B	
reports/review-schematic.json	4,062 B	
reports/review-schematic.md	18,880 B	
reports/schematic.pdf	491,501 B	
reports/top.net	251,873 B	